Temporal Embeddings of Scholarly Mobility

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Abstract

Many researchers name more than one institution in the same year, and each of them ties those institutions together. We collect these ties from OpenAlex for every year from 2008 to 2025 and train one map of institutions per year, in which institutions that share many researchers sit close together. A country sits at the weighted average position of its institutions. Within each year we measure how far a country stands from the rest of the world and how closely two countries lean the same way. A twin world with shuffled years removes the change that comes from growth, and two random halves of the authors give the size of the noise. Net of growth, the average country stands a little farther from the rest from 2019 onwards. Beyond this common drift, 32% of country-years lie more than two noise standard deviations from the country's own average year, against 5% expected from noise. Most countries fall into two large groups, one moving away from the rest and one moving towards it. The maps also show regions leaning towards neighbours that share their language, and Russia moving away from the rest after its invasion of Ukraine. Set against fifty-nine published conclusions about these ties, the maps agree with thirty-two and disagree with twelve.

1 The idea

We map which research institutions work closely together, and how their closeness changes from year to year, through the researchers they share. A researcher who names two institutions in the same year is shared by both, and many shared people point to close work. Typical cases are a professor with a joint appointment, a postdoc who moves halfway through the year and a clinician who also teaches at a university.

> **Example.** In 2019 Mei publishes three papers. Paper 1 lists Kyoto University and RIKEN, paper 2 lists Kyoto and paper 3 lists RIKEN, so her year ties Kyoto to RIKEN.

The papers come from the 2026-09-23 release of OpenAlex, an open index of the scientific literature (Priem et al., 2022), and were published from 2008 to 2025. Murray et al. (2023) used a small neural network to map where scientists move. We train one such network for every year and compare the years.

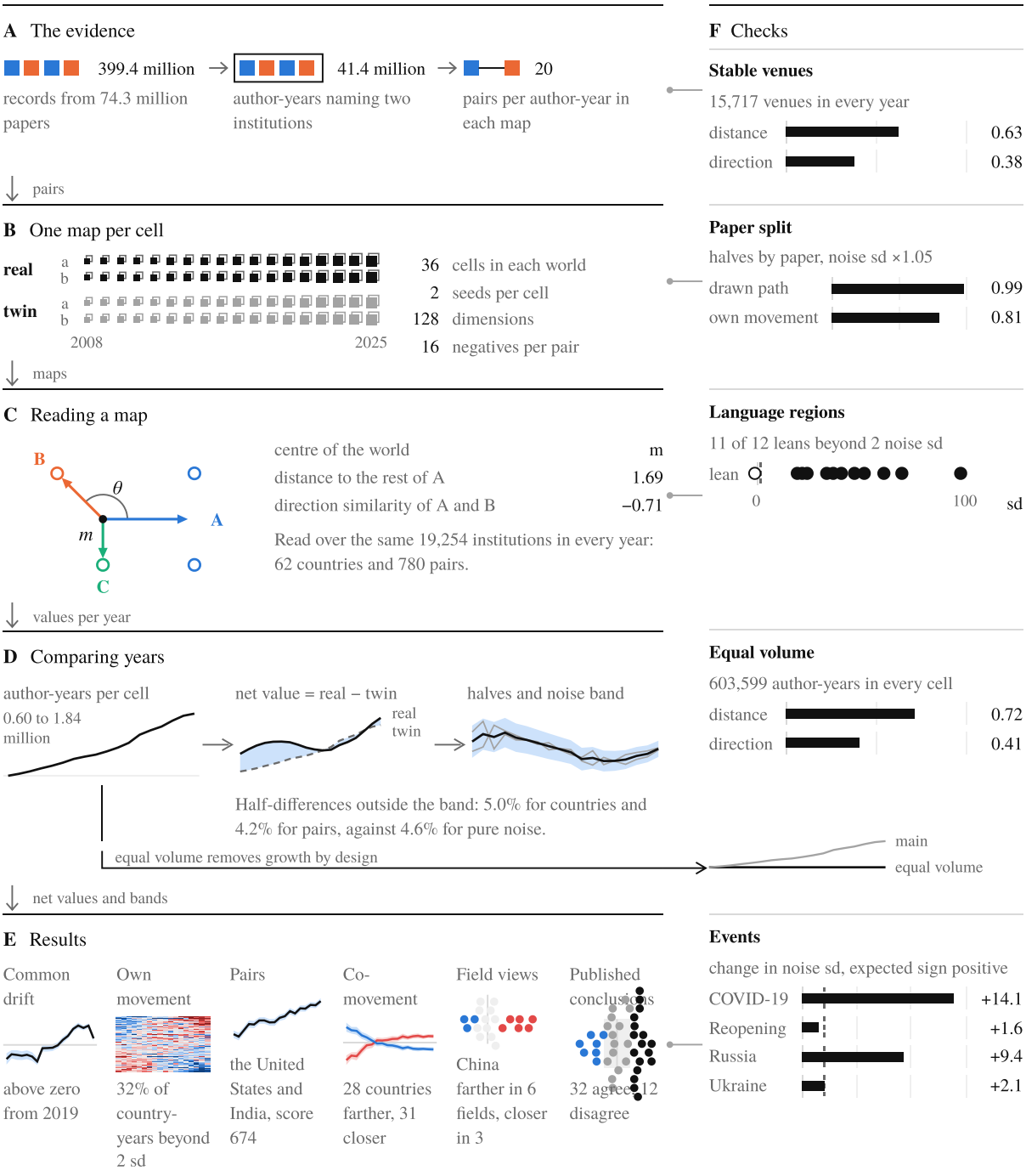


Figure 1. The method on one page. (A) A researcher who names two institutions in one year ties them together, and each such author-year gives 20 training pairs to its map. (B) Every cell, one year of one half of the authors, trains its own map from two seeds, and so does every cell of the twin world, whose years are shuffled. The squares grow with the author-years of each cell. (C) Each map gives every country a distance to the rest and every pair of countries a direction similarity. (D) Data growth moves the raw values of every year. The twin subtracts it after training, and the two halves give the noise band. (E) The results drawn from the data: the common drift, own movement, the leading pair, the two large groups of countries that move together, the change in the own movement of China within each field, and the published conclusions on change that the maps agree with, in ink, or disagree with, in blue. (F) Each check sits beside the step it tests, and its bars give correlations with the main world: of net values for the two check worlds, and of drawn paths and own movement for the paper split. The equal-volume world removes data growth by design, and its net values correlate with those of the main world at 0.72 for distance to the rest and 0.41 for direction similarity. Arrows lead from each step to the next, and the lines in C and D are sketches. Panels A to E follow Section 2 to Section 6, and panel F follows Section 7.

2 Training pairs

2.1 Records and author-years

A **record** is one paper, one author and one institution, and an **author-year** collects all records of one author in one year. An author-year enters the maps when its records name at least two institutions, which a single paper with two affiliations already achieves. It then brings all its records, including repeated ones at the same institution (Figure 2).

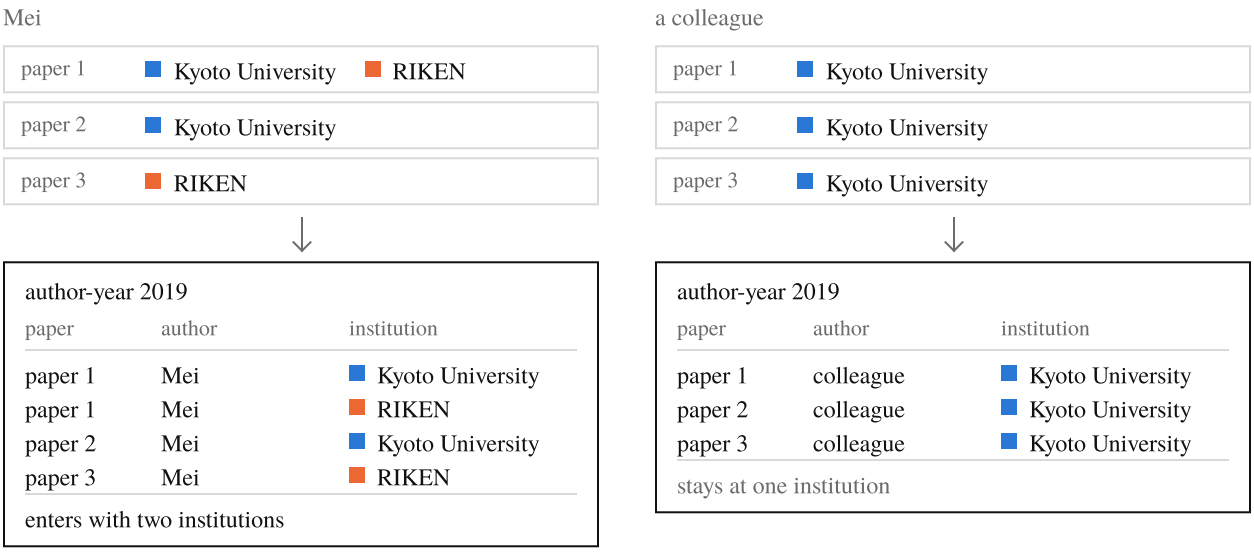


Figure 2. Mei's 2019 as records, one for each pair of paper and institution. Her records name two institutions, so her author-year enters the maps. All three papers of her colleague name Kyoto University alone.

The selection holds 74.3 million papers of six types and 399.4 million records. The 41.4 million author-years that enter the maps carry 276.3 million of these records (Table S5).

2.2 Pairs from each author-year

The network learns from pairs of institutions. To draw one from an author-year, we pick a record at random and then a second among the rest, and two records at the same institution can also form a pair. A balanced split therefore ties two institutions more strongly than a lopsided one with the same number of records (Figure 3).

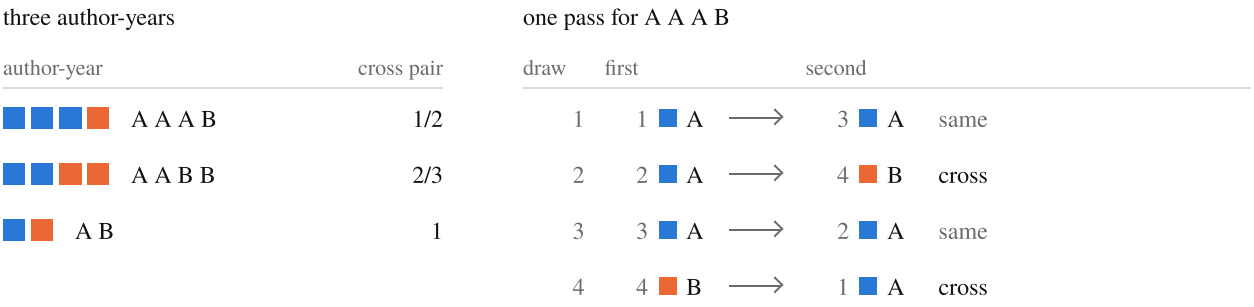


Figure 3. Drawing pairs from one author-year. The first record is drawn among all records and the second among the rest. For the author-years A A A B, A A B B and A B, a draw joins two institutions with probability 1/2, 2/3 and 1.

Every author-year gives four pairs in each pass over the data and 20 over the five passes of training. A researcher with forty records in a year therefore counts as much as one with two (Figure S1). Pairs in proportion to records would give a growing weight to author-years with many records, whose share rises over the years (Section S1).

3 Yearly maps

3.1 Pairs become a map

Every institution starts as a point at a random place, and the network then reads the pairs one at a time. Each pair pulls its two institutions a little closer and pushes one of them away from sixteen institutions drawn at random, with busy institutions drawn more often. After many pairs, institutions that share researchers sit close together, as do two institutions that share many people with a third, and the rest spread out.

The method is skip-gram with negative sampling, first used to place words that occur in similar contexts (Mikolov et al., 2013). Each institution becomes an arrow with 128 coordinates (Table S8, Section S1), and the closeness of two institutions is the cosine of the angle between their arrows. The cosine is 1 for arrows that point the same way, 0 at a right angle and −1 for opposite arrows. The example in Figure 4 uses two dimensions.

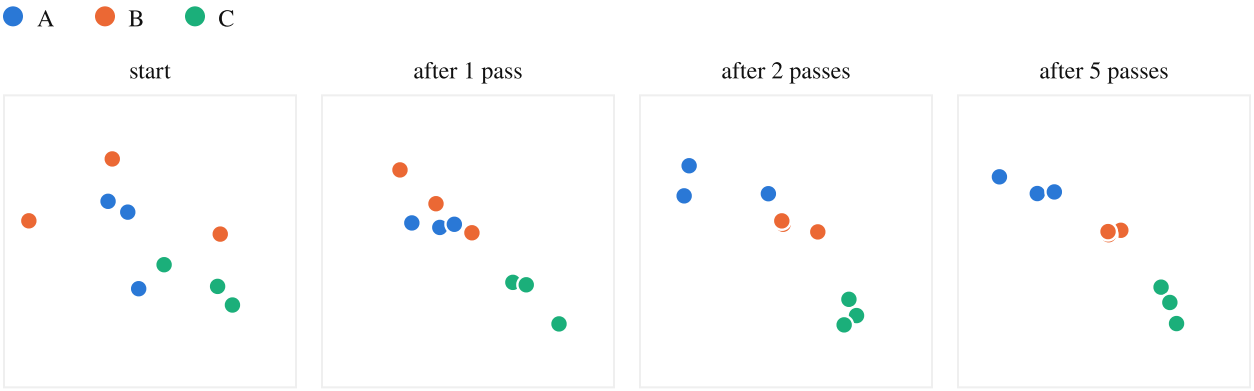


Figure 4. A map forming. Nine made-up institutions in three countries start at random places. Most ties stay inside a country, and B shares far more ties with A and with C than A and C share with each other, so B settles between them.

3.2 Halves and cells

To measure how much chance moves the numbers, we split all authors once into two random **halves**, and each author keeps the same half in all years, so every author-year, the unit of evidence, stays whole in one half (Section S1). One year of one half is a **cell**, and the 18 years give 36 cells (Table S7). Each cell gets its own map, trained twice from different random starts, and every number read from a cell averages its two runs.

4 Reading one map

4.1 The same institutions every year

Each year's map is read over the same 19,254 institutions, those with a known country and at least three records in every cell, both in the data and in the shuffled twin of Section 5 (Section S4, Table S4). Every author-year carries a total weight of one, split over its institutions in proportion to its records, and an institution's **weight** is its share of these totals over all years. Weights stay fixed across years, so any movement of a country on the map comes from its institutions.

4.2 Scale, centre and scale again

Before measuring, we prepare the arrows in three moves (Figure 5, Section S1). We first scale every arrow to length one, since its length mostly tracks how often its institution appears. All arrows also lean towards one shared direction, which pushes every cosine towards one, so we then subtract their weighted average and scale every arrow to length one again. The **centre of the world** is the weighted average of the prepared arrows and lies close to the origin.

Among the 3,000 institutions with the largest weight, a cosine higher by 0.1 on the prepared arrows multiplies the author-years two institutions share, relative to what their sizes predict, by about 2.2 (Section S10).

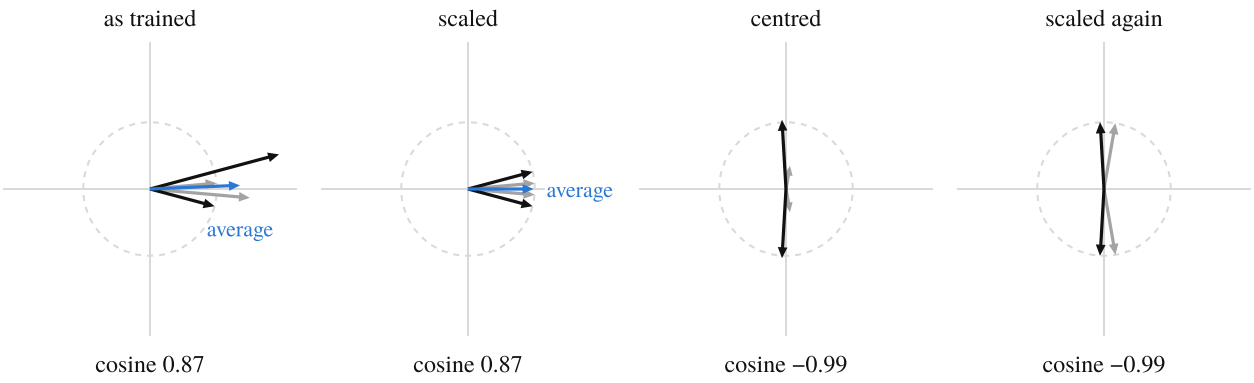


Figure 5. Scale, centre and scale again on four example arrows. The outer two have a cosine of 0.87, and after centring they point in nearly opposite directions, with a cosine of −0.99. Scaling first removes the pull of the longest arrow on the average. The centred arrows are drawn four times longer.

4.3 A country is an arrow

A country sits at its **centroid**, the weighted average position of its institutions, and we draw it as an arrow from the centre of the world to that centroid.

> **Example.** The example map in Figure 6 has three countries and uses raw positions to keep the arithmetic short. Country A has institutions at $(2, 1)$ and $(2, -1)$, each with weight 0.1, so its centroid is $(2, 0)$. Country B is one institution at $(-1, 1)$ and country C is one institution at $(0, -1)$, each with weight 0.4. The weighted average of all four institutions is $(0, 0)$, the centre of the world.

4.4 Distance to the rest

Distance to the rest is the gap between a country's centroid and the centroid of all other institutions, divided by the world's **spread**, the typical distance of an institution from the centre, which stays close to one because every prepared arrow has length one.

$$D_c = \frac{\|g_c - g_{\bar{c}}\|}{\sqrt{S}}, \qquad S = \sum_i w_i \|x_i - m\|^2.$$

Here g_c is the country's centroid, $g_{\bar{c}}$ the centroid of all other institutions, x_i and w_i the position and weight of institution i, and m the centre. The centre lies on the line between the two centroids, so with W_c the country's total weight the gap equals the length of its arrow divided by $1 - W_c$, the weight of the rest.

$$D_c = \frac{\|g_c - m\|}{(1 - W_c)\sqrt{S}}.$$

> **Example.** Country A's arrow has length 2 and A carries a fifth of the weight, so its gap to the rest is $2/0.8 = 2.5$. The spread is $\sqrt{2.2} = 1.48$, which gives a distance to the rest of 1.69.

4.5 Direction similarity

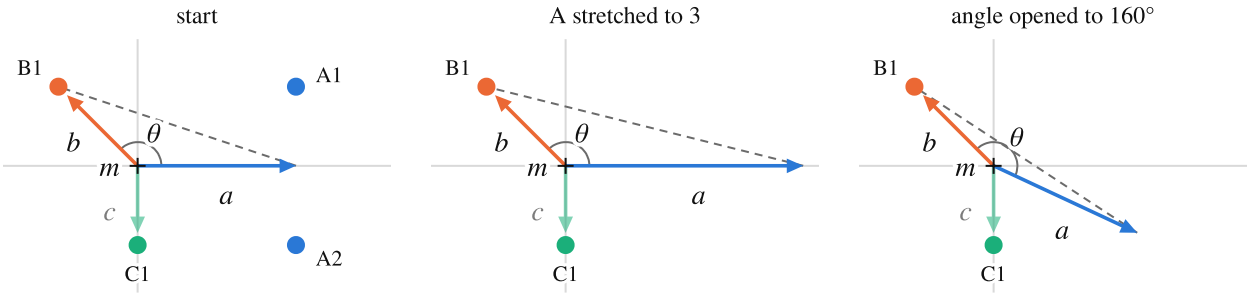
Direction similarity is the cosine of the angle between two countries' arrows.

> **Example.** Country A's arrow is $(2, 0)$ and country C's is $(0, -1)$, so they stand at a right angle with cosine 0. Country B's arrow $(-1, 1)$ stands at 135 degrees from A's, with cosine −0.71.

The plain distance between two countries mixes the two arrow lengths with the angle between them, so a pair can drift apart because one arrow grows or because the angle opens. Distance to the rest reads the length of each arrow and

direction similarity reads the angle, which tells the two cases apart (Figure 6). With a and b the arrows of two countries, the distance between their centroids is $\|a - b\|$, and

$$\|a - b\|^2 = \|a\|^2 + \|b\|^2 - 2\|a\|\|b\| \cos\theta.$$

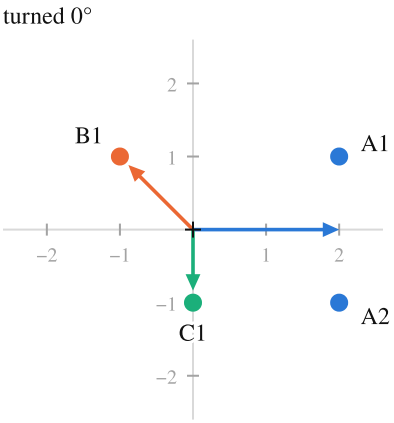


$\|a\|$	2.00	$\|a\|$	3.00	$\|a\|$	2.00
$\|b\|$	1.41	$\|b\|$	1.41	$\|b\|$	1.41
$\cos\theta$	−0.71	$\cos\theta$	−0.71	$\cos\theta$	−0.94
$\|a\|^2$	4.00	$\|a\|^2$	9.00	$\|a\|^2$	4.00
$\|b\|^2$	2.00	$\|b\|^2$	2.00	$\|b\|^2$	2.00
$-2\|a\|\|b\|\cos\theta$	4.00	$-2\|a\|\|b\|\cos\theta$	6.00	$-2\|a\|\|b\|\cos\theta$	5.32
$\|a-b\|^2$	10.00	$\|a-b\|^2$	17.00	$\|a-b\|^2$	11.32

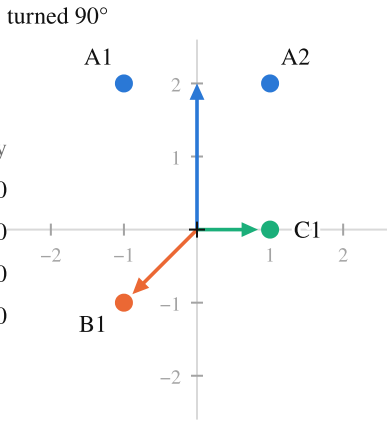
Figure 6. The example map as arrows from the centre of the world m. Stretching A to length 3 raises the squared distance between A and B from 10 to 17 at the same cosine. Opening the angle θ to 160 degrees raises it to 11.32 at the same lengths.

4.6 Turning a map

Turning a whole map keeps every distance and angle. A turned copy therefore fits the data equally well, and each training run lands at its own angle with its own coordinates. Both measures are built from distances and angles inside one map, so they keep their values under any turn and compare across runs and years (Figure 7).



	x	y
A1	2.00	1.00
A2	2.00	−1.00
B1	−1.00	1.00
C1	0.00	−1.00



	x	y
A1	−1.00	2.00
A2	1.00	2.00
B1	−1.00	−1.00
C1	1.00	0.00

distance to the rest	A 1.69	B 1.59	C 1.12
direction similarity	A–B −0.71	A–C 0.00	B–C −0.71

Figure 7. The example map turned. Every coordinate changes, and the distances to the rest and the direction similarities keep their values.

5 Comparing years

5.1 Raw numbers move with the data

A country's distance can rise from one year to the next because the country moved or because the data changed. The cells grow from 603,599 author-years in 2008 to 1,843,864 in 2025, new institutions appear, and larger training sets give sharper maps (Figure S3). Each of these changes can shift every number while the pattern of ties stays the same, so we compare against a world with the same growth and mixed timing.

5.2 The shuffled twin

The **twin world** holds the same data with shuffled years. Each author-year gets a **main institution**, the one with most of its records, and a random pick settles equal counts. Within each half and each main institution we shuffle the years of the author-years, each moving whole with its records, so the twin keeps the number of author-years of every main institution in each year and mixes the timing of its ties (Figure 8, Section S5). We train and read the twin like the real world.

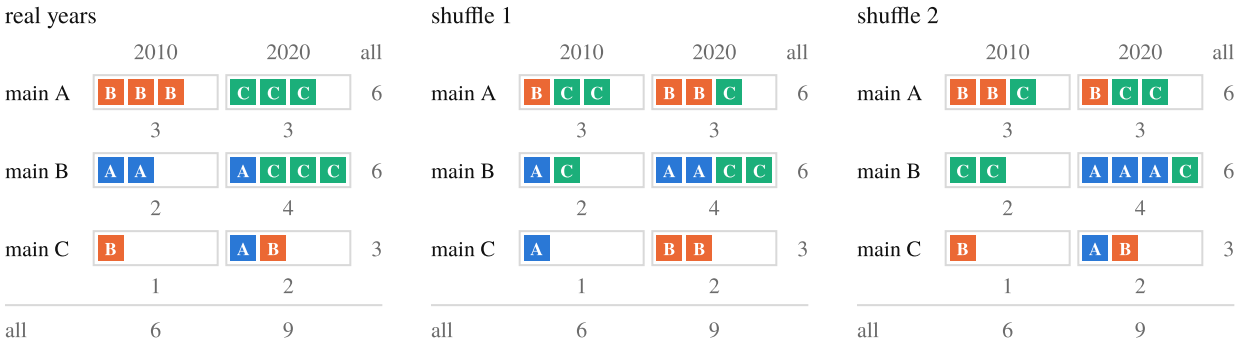


Figure 8. The shuffled twin. Each card is one author-year of a main institution, coloured by its partner. Shuffling moves cards between years within a row, and the counts under each year stay fixed.

5.3 Net values at the real level

The **net value** of a year is the real value minus the twin's. Growth moves both worlds alike, so the rise and fall of the net value shows change in ties. Because the twin keeps most of the structure of the map, the net value sits near zero, and we draw each line at the real average with the shape of the net value. Figure 9 takes China through these steps. Over all countries, the year-to-year shapes of the real and the twin distance correlate at 0.82, so most of the raw movement follows the data.

The map viewer draws country paths from the same maps (Section S9). Figure 10 takes the thirty countries with the most institutions from maps at their own angle to the paths of the viewer. A country moves 0.13 a year on average while each map keeps its own angle, 0.019 once the maps are aligned and 0.016 in the twin world.

5.4 Noise from two halves

The two halves measure the same world with different people, and they differ by chance in which researchers they hold and in how training went. Their average carries the signal plus noise. In half their difference the signal cancels and noise of the same typical size remains (Figure 11).

Each country and each pair has a single half-difference per year, so we pool the years and model the noise standard deviation (sd) of unit u in year t as $\mathrm{sd}_{u,t} = s_u P_t$, one size per unit times one curve over the years, with one curve for all countries and another for all pairs. Small countries get larger sizes, and the curve is higher in the early years, when cells are smaller. The **noise band** runs two noise sd above and below each line, and pure noise leaves it in about one year in twenty (Section S6, Table S13). Splitting the halves by paper in place of author gives noise of about the same size (Table S1).

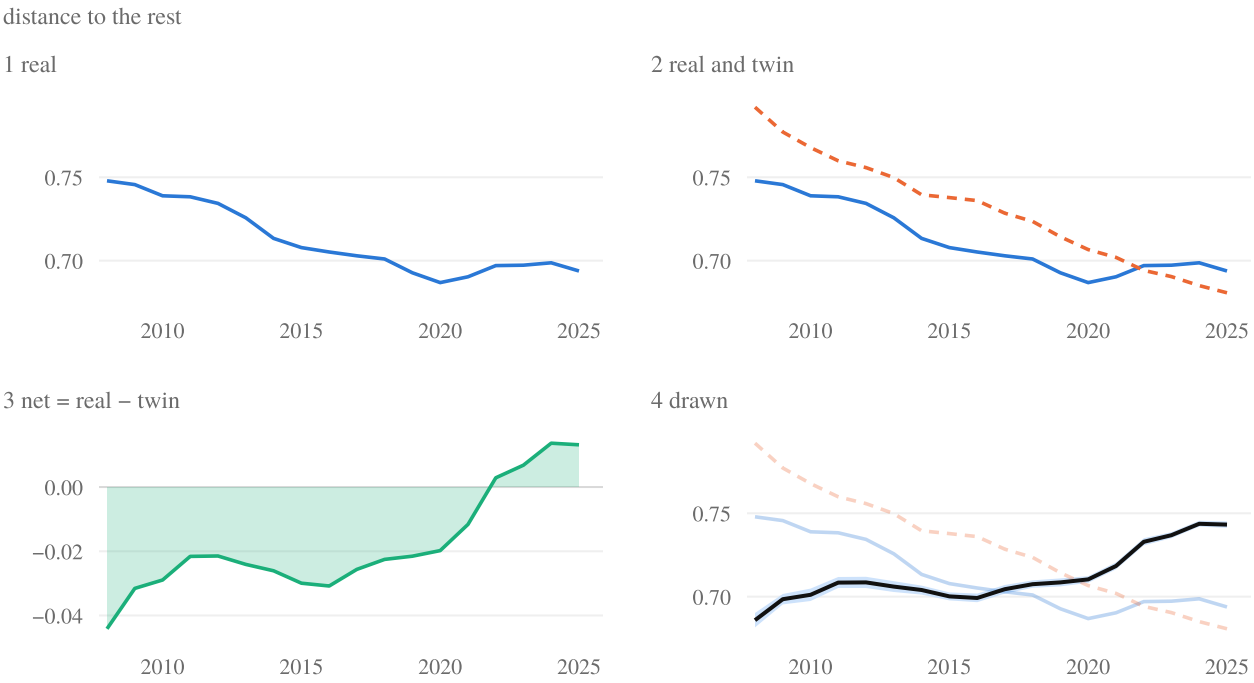


Figure 9. From the real world to a drawn path, for China. Step 1 gives the real distance to the rest in each year, and step 2 adds the twin, dashed. Step 3 gives the net value, the real value minus the twin's, in the lower panel. Step 4 centres the net value and lifts it to the real average with its noise band, as each line of Figure 15 is drawn. The printed figure shows the four steps side by side.

5.5 Common drift and own movement

A country's **net movement** in a year is its net value minus its average net value over all years. The **common drift** is the average net movement of the 62 countries, each counted equally. A country's **own movement** subtracts the drift and divides by the country's noise sd, so a value of +3 puts the country three noise sd farther from the rest than in its average year, beyond the common drift.

> **Example.** In one year, three countries stand +0.010, +0.006 and +0.002 above their average year. The drift is +0.006, so the differences are +0.004, 0 and −0.004. With a noise sd of 0.002, the own movements are +2, 0 and −2.

5.6 How much a pair moved

A pair's **movement score** adds up its squared yearly net values, each centred over the years and measured in noise sd, and keeps any change shared by all pairs. Pure noise gives about 17, one less than the number of years, and larger scores point to movement beyond noise.

> **Example.** Three years at +1, −1 and 0 noise sd give a score of 2, and three years at +4, 0 and −4 give 32.

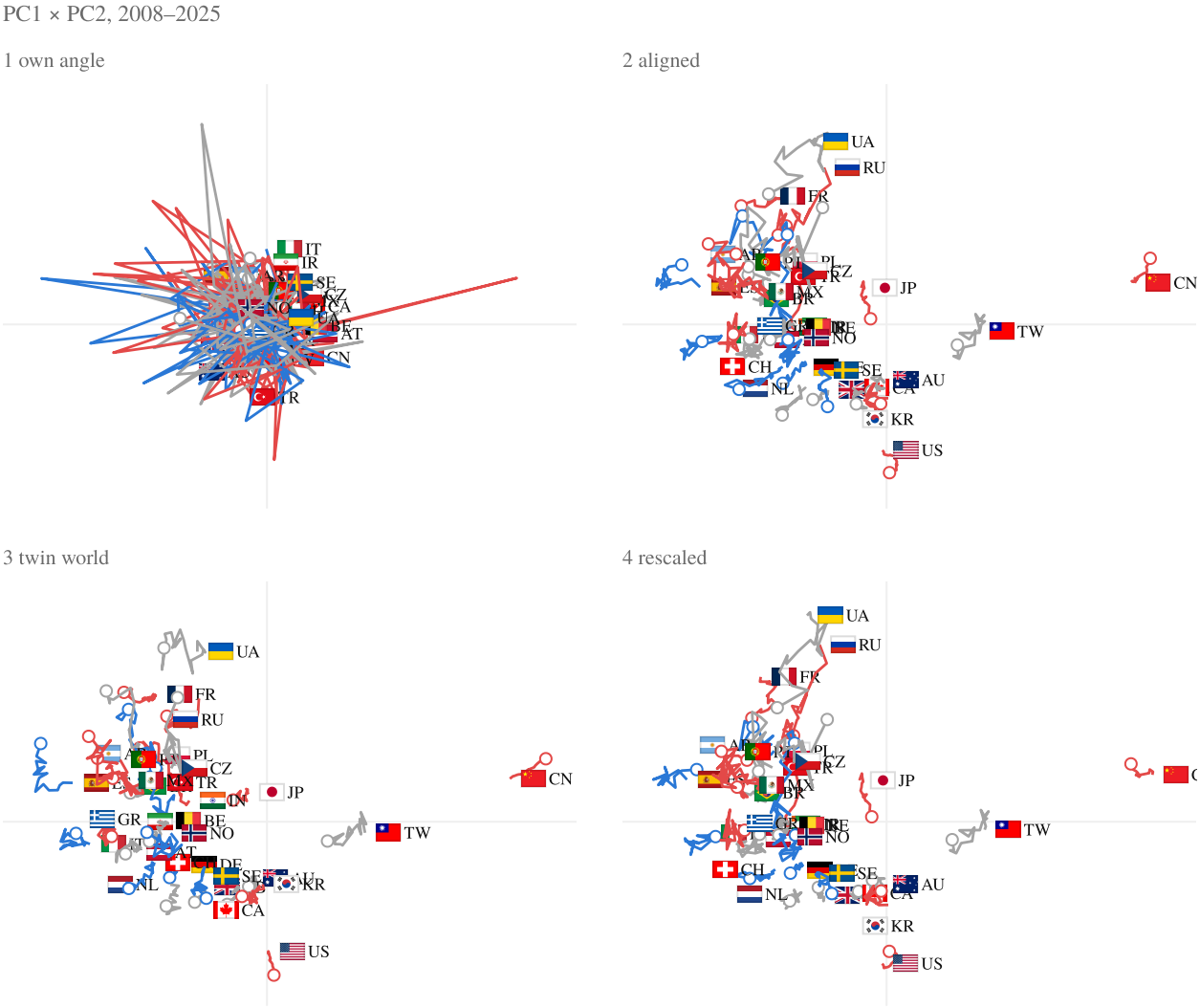


Figure 10. Country paths of the map viewer for the thirty countries with the most institutions. Each point is the weighted centroid of a country's institutions in one year, on the first two principal components of the aligned maps of all years together. Step 1 draws one map per year, from one half and one seed, at the angle its training left it (Section 4). Step 2 turns all maps of all years onto one common reference by a weighted Procrustes alignment and averages each year over halves and seeds. Step 3 turns the maps of the twin world onto the same reference and draws them on the same components (Section 5). Step 4 takes the positions of step 2 and, in each year, multiplies each component by the factor that brings the twin's spread along it to its average over all years, which gives the paths of the viewer (Section S9). An open circle marks 2008 and the flag sits at 2025. A path is red when the country's own movement rises beyond the noise band from 2008–2010 to 2023–2025, blue when it falls beyond it, and grey otherwise. The printed figure shows the four steps side by side.

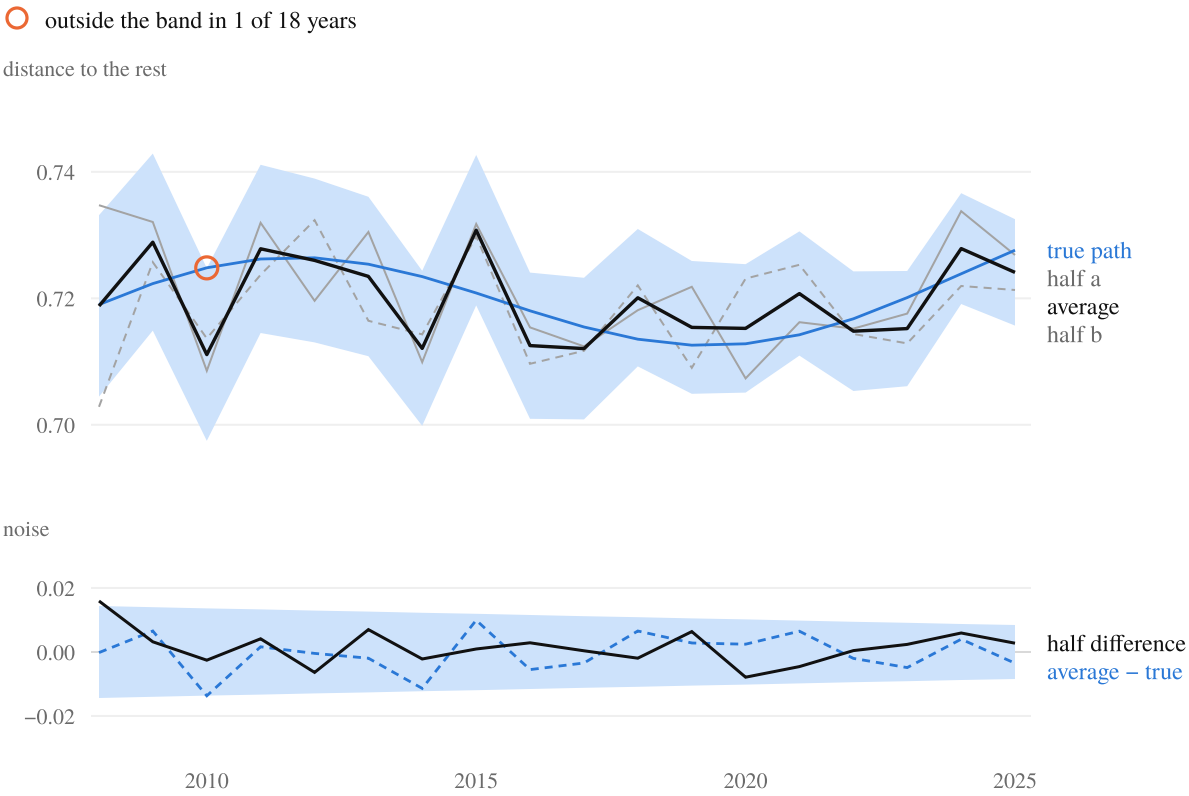


Figure 11. Noise from two halves, on a made-up series. The upper panel shows the true path, both halves and their average with its noise band. The lower panel sets half the difference of the halves beside the error of the average.

6 What the maps show

We read distance to the rest for the 62 countries with at least 20 institutions and direction similarity for the 780 pairs among the 40 countries with the largest total weight. The pairs also include Hong Kong, which has fewer institutions than a distance series needs.

6.1 Where countries stand

Averaged over the eighteen years, the United States stands closest to the rest of the world at 0.427, followed by the United Kingdom and Canada, and Argentina is farthest at 0.891 (Figure 12, Table S14).

6.2 How countries moved

Net of growth, the common drift is negative up to 2018 and positive from 2019, when the average country starts to stand farther from the rest than in its average year. It rises by about 0.02 from its low in 2014 to its high in 2022 (Figure 13, Table S15).

In 32% of the country-years in Figure 14, a country stands more than two noise sd from its average year beyond the common drift, against about 5% for noise alone. China, whose noise is the smallest of any country, moves away from the rest from 2022 onwards and reaches 28 noise sd beyond the drift in 2025, about 0.028 in distance to the rest. For scale, Russia moved 0.017 farther from the rest after 2022 and Ukraine 0.009 closer. The common drift rose by 0.009 after 2020 (Section S14).

Figure 15 follows the eight countries with the largest sums of squared own movement, and Section S12 splits their change and that of the leading pairs over institutions.

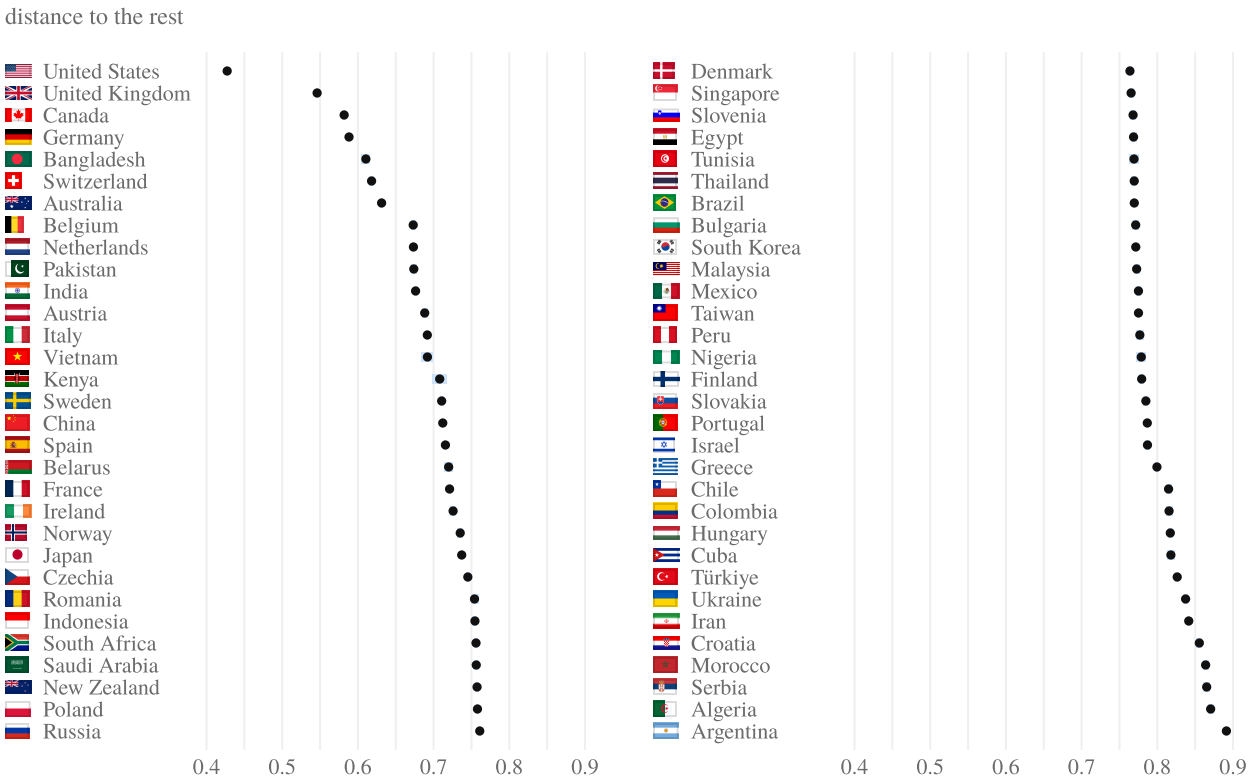


Figure 12. Each country's mean distance to the rest over 2008–2025, in units of the spread. Bars span two noise sd of the mean on either side.

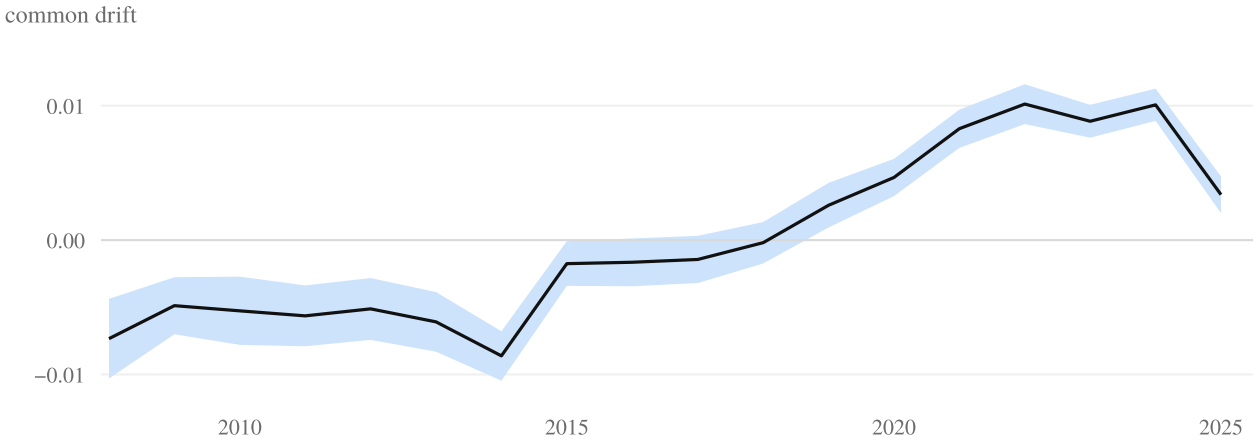


Figure 13. The common drift, the average net movement of all 62 countries, in units of the spread, with its noise band.

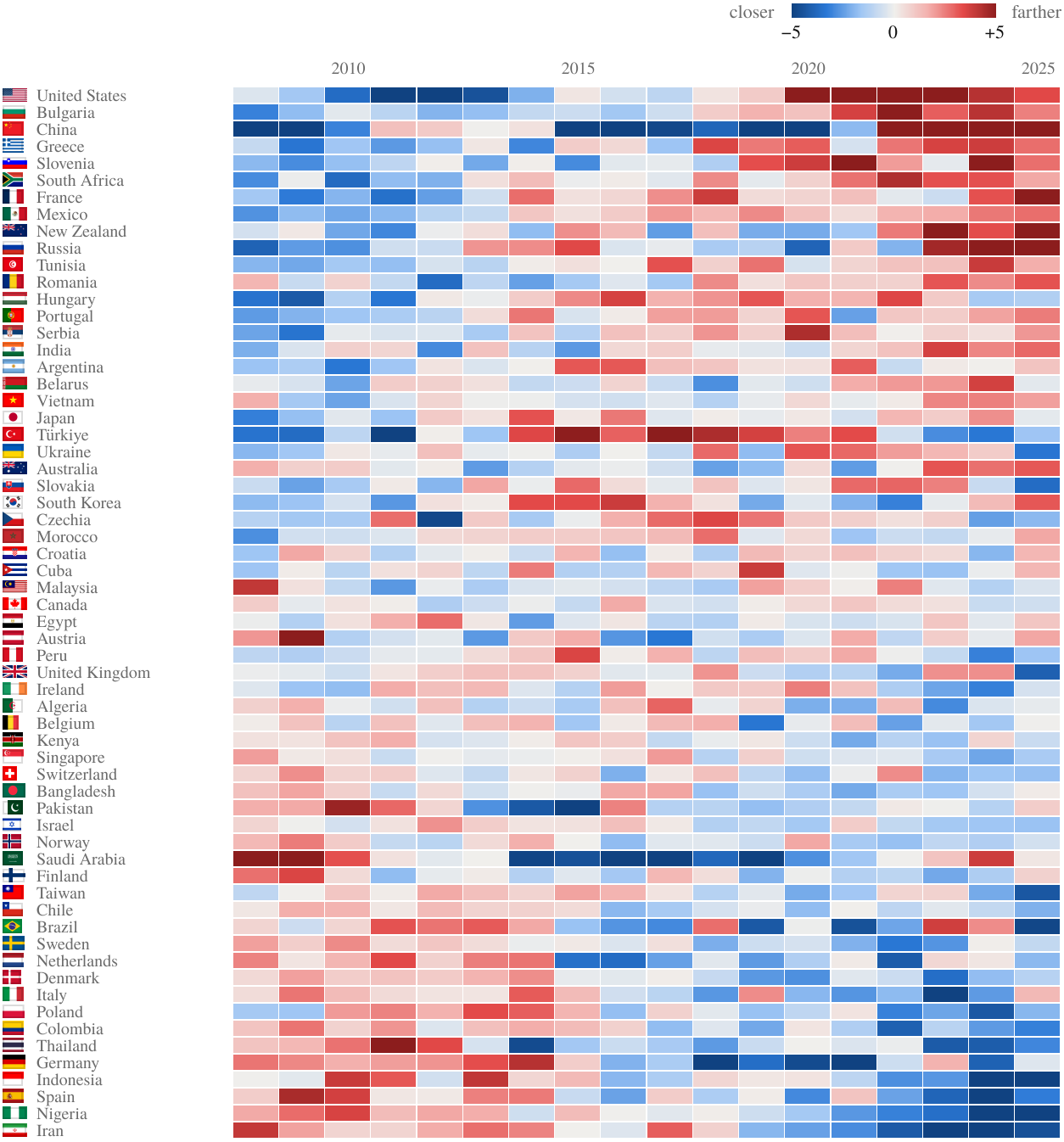


Figure 14. Own movement of all 62 countries, in units of each country's noise sd. Red years stand farther from the rest than the country's average year, beyond the common drift, and blue years closer, with colours saturating at ±5. Countries that move away from the rest in the later years sit at the top and those that move closer at the bottom.

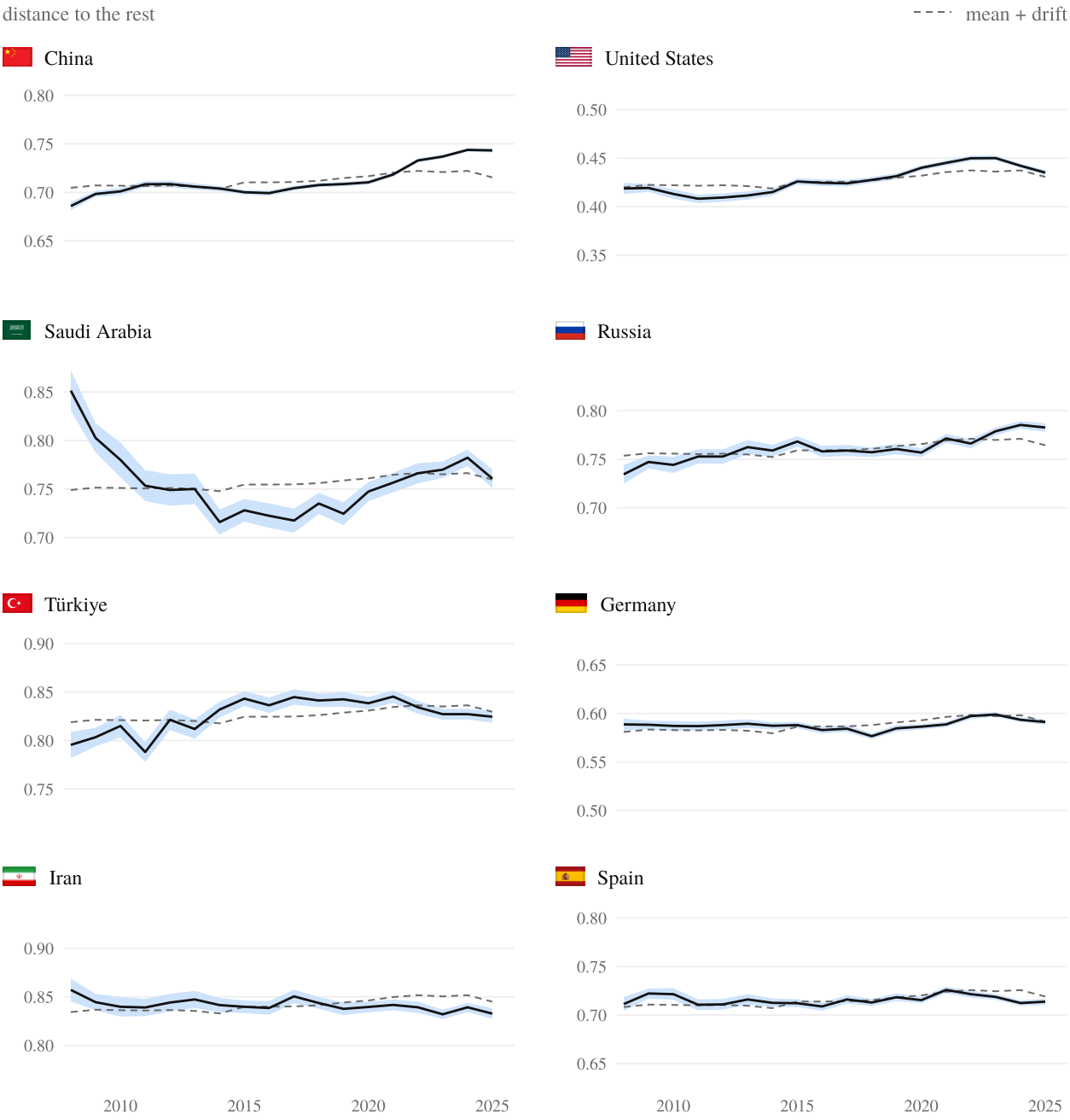


Figure 15. The eight countries with the largest sums of squared own movement. Each panel shows a country's net distance to the rest, drawn at its real average as in Figure 9, with its noise band and, dashed, the path that the common drift alone would give. Every panel spans the same range of distance.

6.3 Countries that move together

Countries also move in groups. We correlate the movement of each country beyond the common drift in one half with that of every other country in the other half, and cluster the countries on these correlations (Section S11). Clustered separately, the two halves agree best on three groups, with an adjusted Rand index of 0.52. Clusters of the 780 pairs reach at most 0.12, so co-movement is reported for countries alone.

Between 2008–2010 and 2023–2025, a group of 28 countries whose largest members are China, the United States and France moves 15.9 noise sd farther from the rest (Figure 16). A group of 31, whose largest members are the United Kingdom, Germany and Brazil, moves 15.6 noise sd closer. Movement beyond the drift averages zero over the countries in every year, so the two moves mirror each other, and both groups had covered half of their change by 2013. The third group, India, Canada and Romania, moves 2.0 noise sd farther, at the edge of the band.

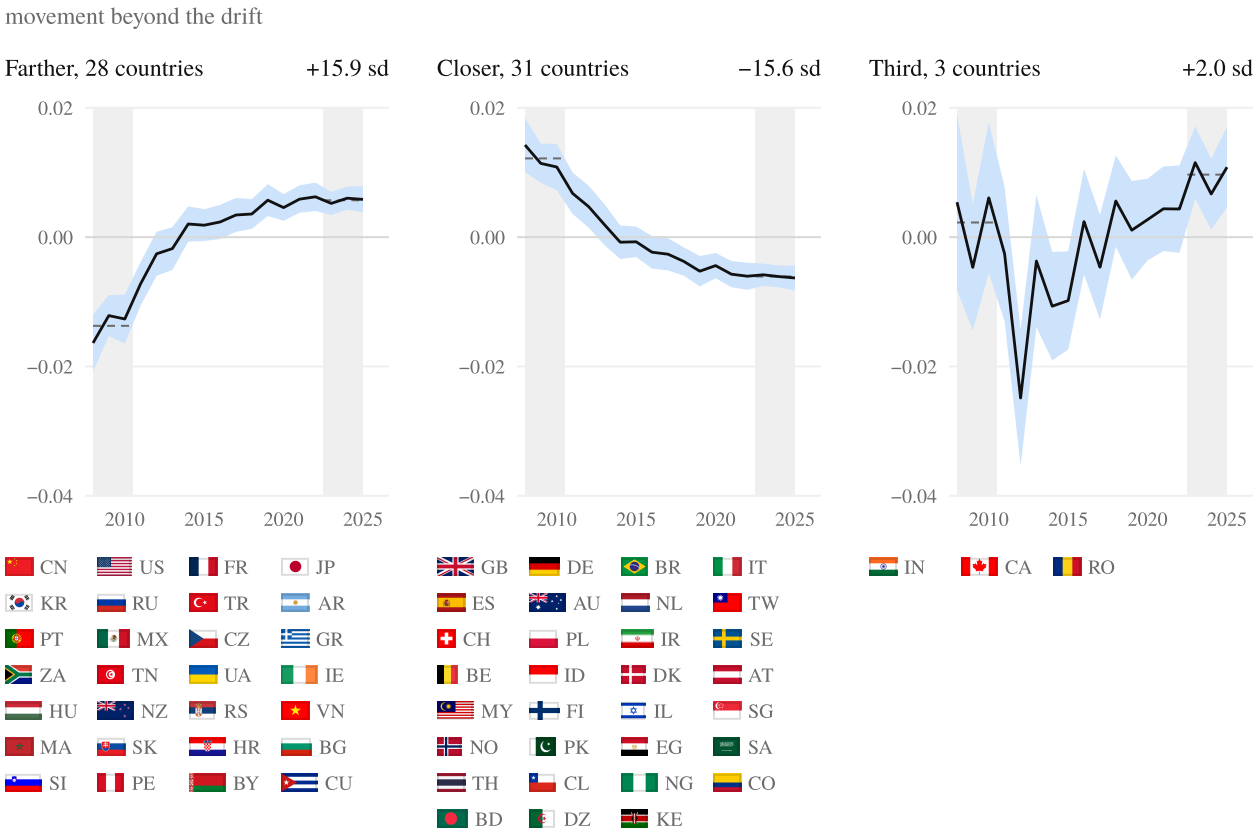


Figure 16. Mean movement beyond the common drift of each group of countries, in units of the spread, with its noise band. All panels share one vertical axis. Shaded columns mark 2008–2010 and 2023–2025 with their means dashed, and the number in each head gives the change between them in noise sd.

6.4 How pairs of countries lean

The mean direction similarity over the years runs from −0.581 for China and Spain to 0.605 for Brazil and Portugal (Figure 17).

The median pair scores 25 against about 17 for pure noise. Figure 18 shows the twelve pairs with the largest scores, led by the United States and India at 674, China and Malaysia at 562 and China and Pakistan at 493 (Table S16). Drawn at its real average, the leading pair rises from −0.19 in 2008 to +0.16 in 2025.

6.5 Field views

Field views read the same measures inside each of the 26 OpenAlex fields. Each record takes the subfield of one work its paper cites, drawn at random among the cited works with a topic, and each institution is measured through its records

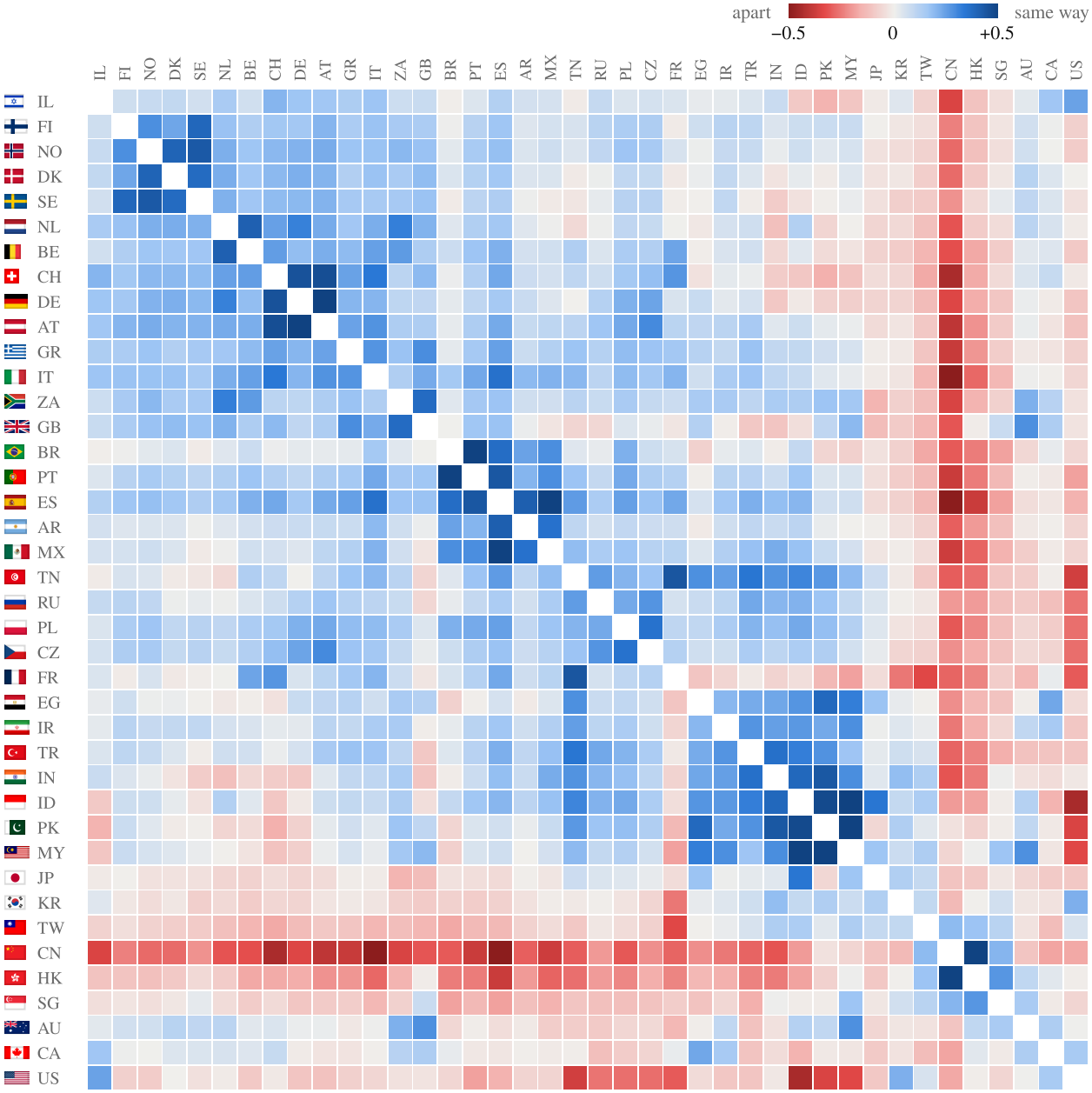


Figure 17. Direction similarity among the 40 countries with the largest total weight, averaged over the years. Rows and columns follow a clustering of the countries by their similarities. Blue pairs lean the same way from the centre and red pairs lean apart, with colours saturating at ±0.5.

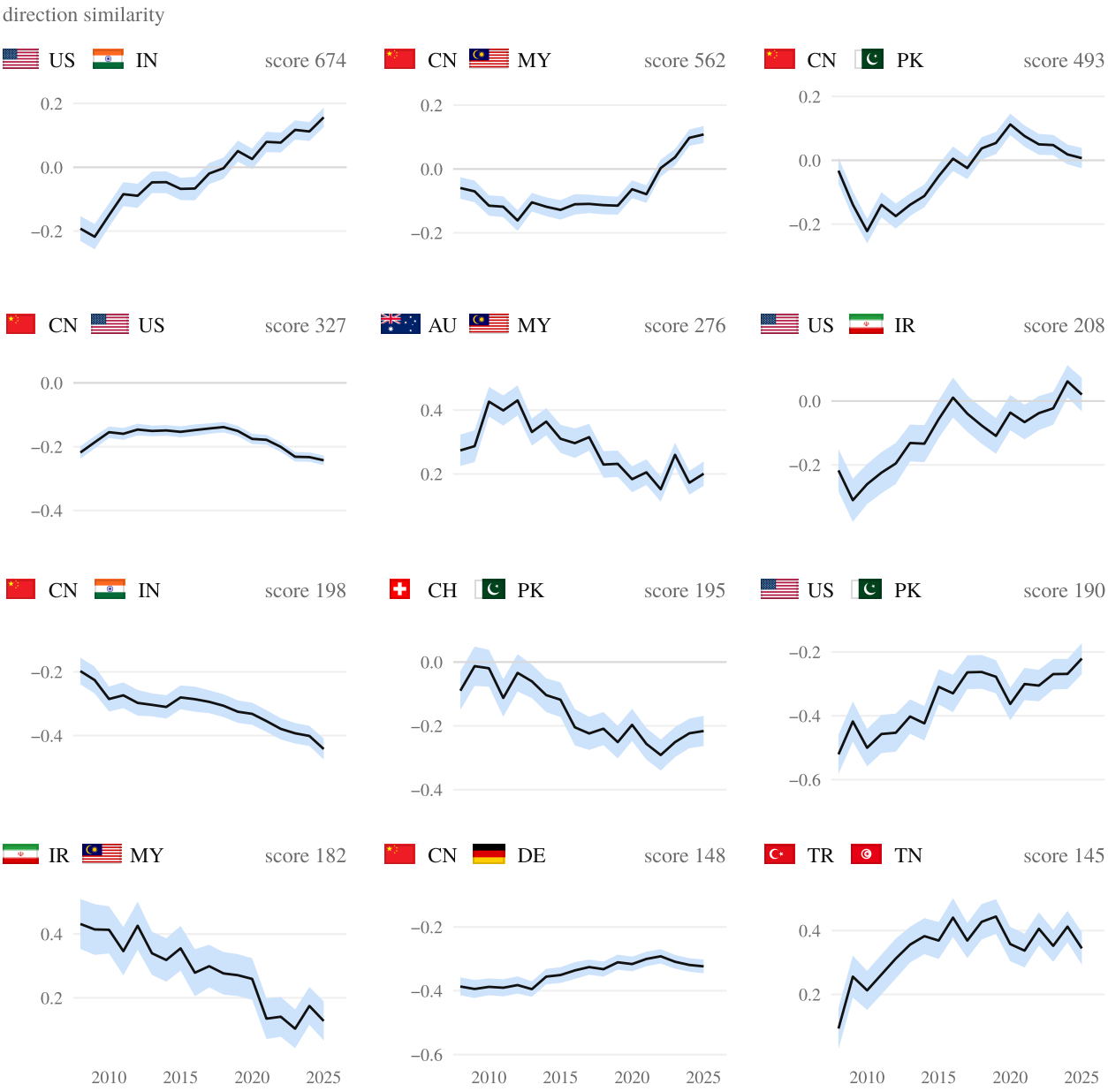


Figure 18. The twelve pairs with the largest movement scores, each with its score. Each panel shows the pair's net direction similarity drawn at its real average, with its noise band, and every panel spans the same range.

in the field. The centred net distances of the two halves then correlate at 0.62, against 0.57 when each record takes the subfield of its paper's primary topic (Section S15 gives the rule, the comparison and the subfields). From 2008–2010 to 2023–2025 the own movement of China rises beyond the noise band in six of the 20 fields where it has a distance series, most in Engineering, and falls beyond it in three, most in Computer Science (Figure 19). In the main world, which pools all fields, it moves away from the rest from 2022 onwards.

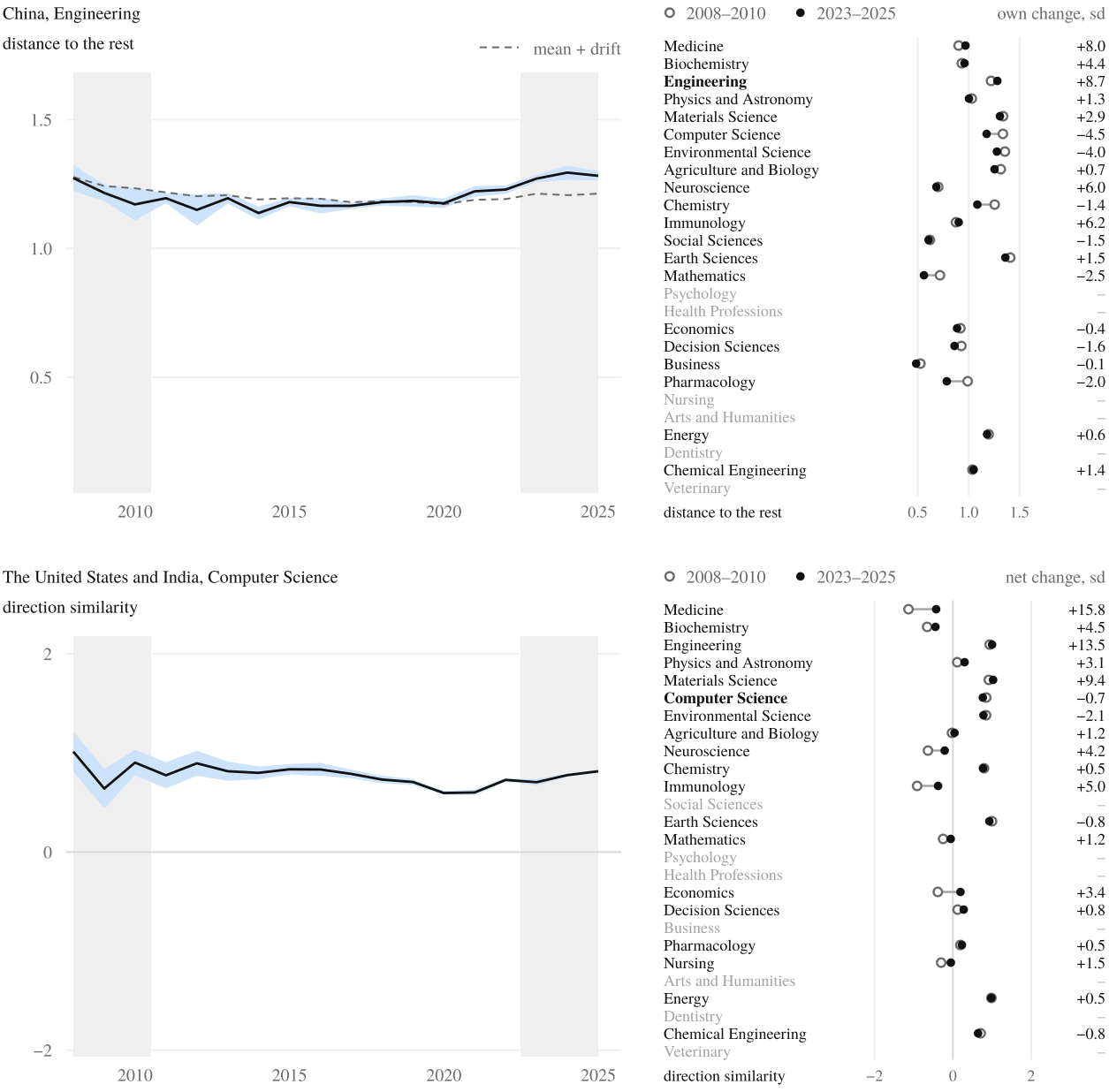


Figure 19. Field views of one country or one pair. The vertical axis gives distance to the rest for a country and direction similarity for a pair, each drawn at its real average with the year-to-year shape of its net value, as in Figure 15, on one scale for all fields. The band spans two noise sd, shading marks 2008–2010 and 2023–2025, and for a country the dashed line adds the common drift of the field to the country's mean. The list on the right shows every field on the same scale, with the mean over 2008–2010 as an open dot and the mean over 2023–2025 as a filled dot, and its last column gives the change between the two periods in noise sd, in own movement for a country and in net value for a pair. The printed figure shows China in Engineering and the United States and India in Computer Science, the country and the pair with the most movement in the main world, each in the field where it moves most.

7 Checks

7.1 Equal volume and stable venues

Two more worlds rerun the whole pipeline of the **main world**, each with its own twin. The **equal-volume world** fixes the volume of data by drawing 603,599 whole author-years for every cell, the size of the smallest cell. The **stable-venue world** keeps the papers of the 15,717 venues that publish in every year from 2008 to 2025, which limits the effect of changes in OpenAlex coverage (Section S8).

The centred net values of the main world correlate with those of the equal-volume world at 0.72 for distance to the rest and 0.41 for direction similarity, and with those of the stable-venue world at 0.63 and 0.38 (Table S17).

7.2 Expected patterns

Shared language and two events give patterns to expect. A part of a country that shares a language with a neighbouring country, or lies on the Franco-German border, should lean towards that neighbour more than the rest of its country does. The lean is the part's direction similarity with the neighbour minus that of the rest of its country, and it passes when its mean over the years lies more than two noise sd above zero (Section S13). Eleven of the twelve leans pass, each at least 19 noise sd above zero, led by Quebec towards France at +0.415 (Figure 20). The German side of the border leans towards France about as much as the rest of Germany does, at −0.007.

The pandemic cut travel and visiting posts from 2020, so the common drift should rise and then fall after the reopening of 2023. After the invasion of Ukraine in 2022, Russia should move farther from the rest and Ukraine closer to it. Each change compares the mean of the three years after the onset year with that of the three years before it, and passes when it lies more than two noise sd away in the expected direction (Section S14). The drift rose by 14.1 noise sd, but it was already rising in the three years before 2020, so part of this rise predates the pandemic. Russia moved 9.4 noise sd farther from the rest and Ukraine 2.1 closer, with most of Ukraine's change in 2025. The fall of 1.6 noise sd after the reopening, over the two years to 2025, stays inside the band.

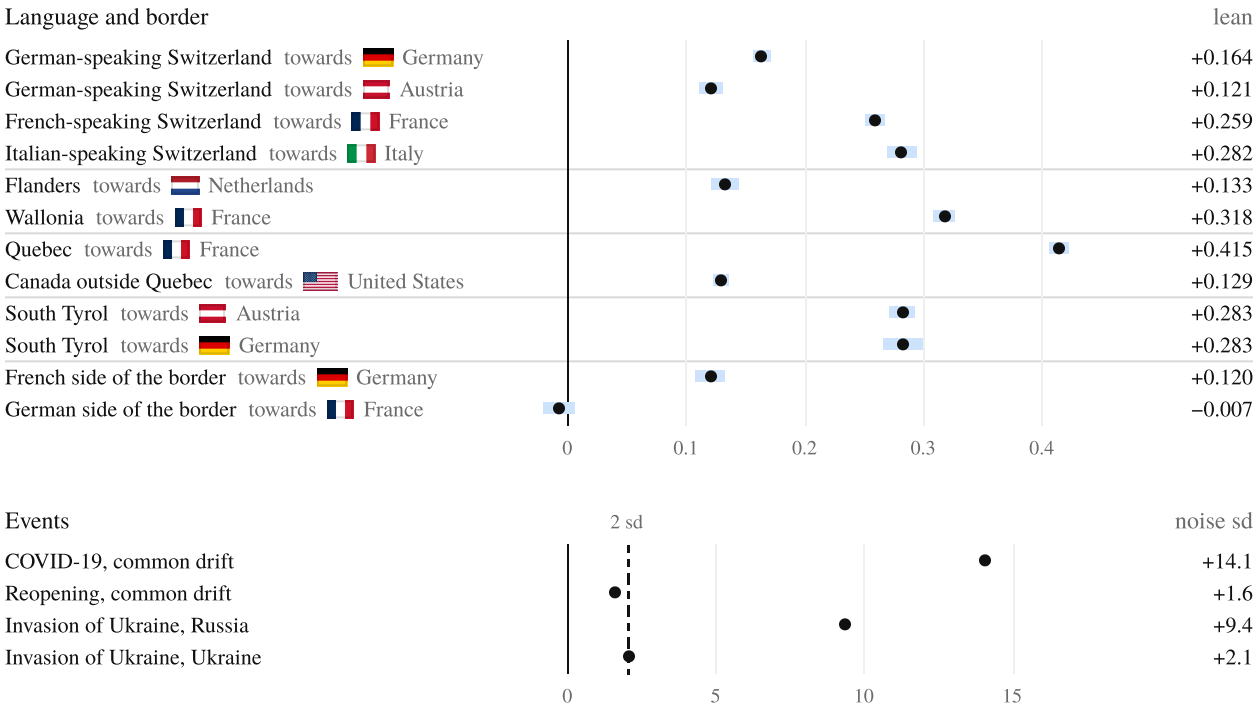


Figure 20. Expected patterns. The upper panel gives the lean of each language part or border side towards its neighbour, averaged over 2008–2025, with a bar of two noise sd of the mean on either side. The lower panel gives the change around each event in noise sd, signed so that the expected direction is positive, with a dashed line at two noise sd.

8 Published conclusions

Studies of international research ties have drawn their conclusions from counts of coauthored papers, from networks built on these counts, from gravity models and from samples of affiliations. We set fifty-nine of these conclusions, listed with their studies in Table 1, against the maps. Each conclusion becomes a change between two windows of years inside the period its study covers, of the common drift, of one country's movement net of that drift or of a pair's direction similarity, in noise sd, or a rank against the median country or pair. A conclusion agrees with the maps when the value lies beyond the noise band in the direction it implies and disagrees when the value lies beyond the band in the other direction, and a conclusion that ties held agrees while the value stays inside the band.

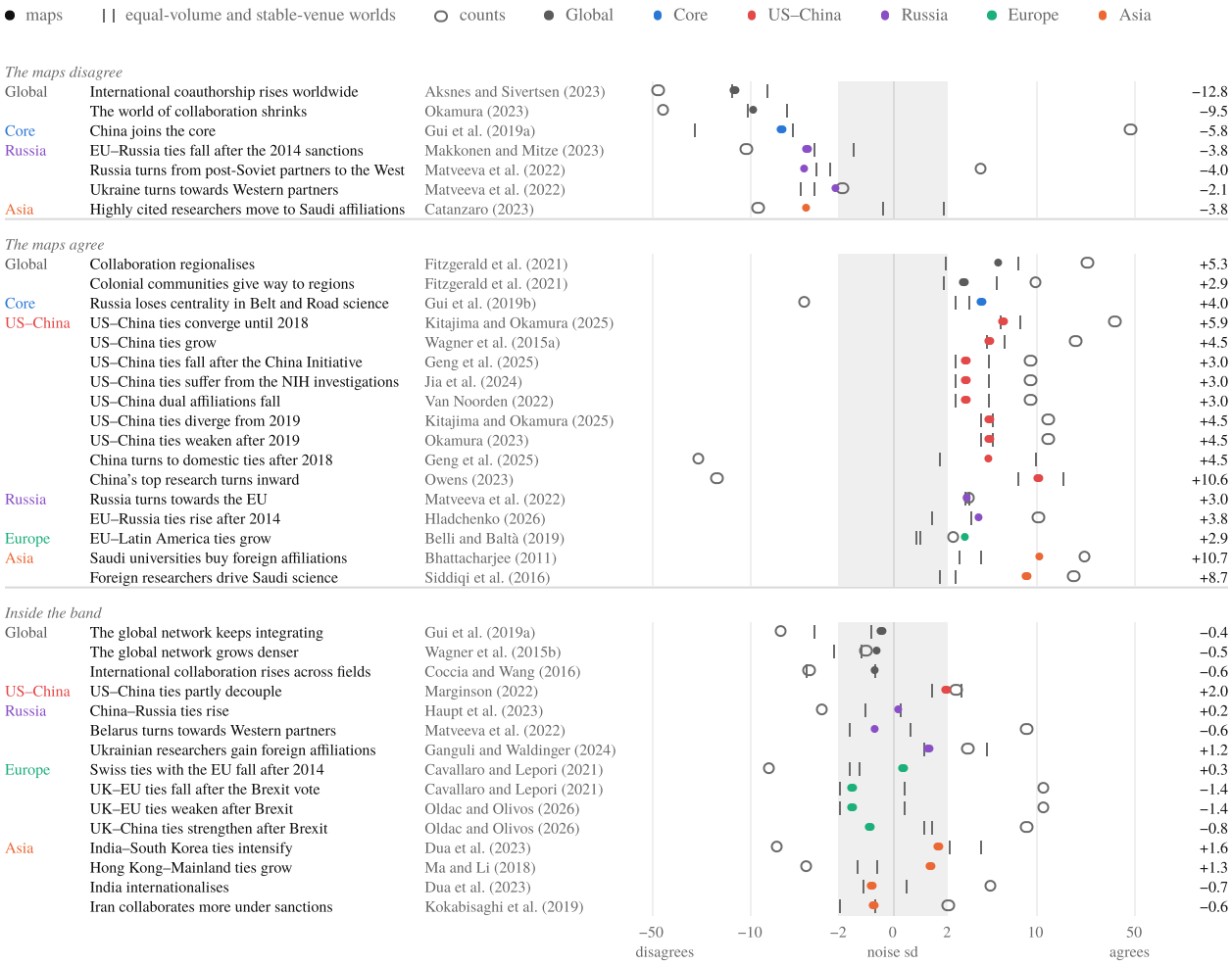


Figure 21. Published conclusions on change against the maps, grouped by verdict and within each verdict by theme, with one colour for each theme. Each row gives one conclusion and its study, and the dot gives the value of the maps in noise sd, signed so that values to the right agree with the conclusion and values to the left disagree. The shaded band spans two noise sd on either side of zero, the ticks give the equal-volume and the stable-venue world, and the open circle gives the counts. Conclusions that ties held and conclusions on levels appear in Table 1 only.

The maps agree with thirty-two conclusions, disagree with twelve and leave fifteen inside the band (Figure 21). They agree with the rise of US–China ties up to 2018 and their fall after it (Geng et al., 2025; Kitajima & Okamura, 2025), with China's turn to domestic ties after 2018 (Geng et al., 2025; Owens, 2023), with the pull of Saudi institutions on foreign researchers after 2011 (Bhattacharjee, 2011; Siddiqi et al., 2016) and with the central place of the United States, the United Kingdom, Germany and China in co-affiliations (Hottenrott et al., 2021).

Two studies find global collaboration drawing together, through a rising share of internationally coauthored papers (Aksnes & Sivertsen, 2023) and a shrinking distance between countries (Okamura, 2023). Net of growth, the average country moves away from the rest over their windows, by 12.8 noise sd up to 2019–2021 and by 9.5 up to 2018–2020. The same study finds US–Japan ties flat over the 2010s, while their arrows turn apart by 3.7 noise sd.

Studies of Russia find its ties turning from post-Soviet partners to the West (Matveeva et al., 2022) and holding after the invasion of Ukraine (Hladchenko, 2026; Zhang et al., 2024). In the maps Russia moves away from the rest by 4.0 noise sd from 2008–2010 to 2016–2018 and by 5.1 after 2022, when the arrows of Russia and the EU turn apart by 4.2 noise sd. Ukraine moves away from the rest by 2.1 noise sd up to 2016–2018, where the first study finds it turning to the West. A gravity model finds EU–Russia ties falling after the sanctions of 2014 (Makkonen & Mitze, 2023), while their arrows turn together by 3.8 noise sd, as relative intensity also shows (Hladchenko, 2026).

Network studies place China in the core of the global network over 2000–2015 (Gui et al., 2019), while China moves away from the rest by 5.8 noise sd from 2008–2010 to 2013–2015. A review places China and India outside the Euro-American core (Marginson, 2022), while they rank 18th and 10th of the sixty-two countries by distance to the rest, nearer than the median country. A news analysis reports highly cited researchers moving their affiliations to Saudi institutions after 2013 (Catanzaro, 2023), while Saudi Arabia moves away from the rest by 3.8 noise sd up to 2021–2023.

Nine of the twelve disagreements hold in both the equal-volume and the stable-venue world. The counts column reads the same records with the measure closest to each study's own, through the same twin and noise model: the international share for a country, the share of its co-affiliated author-years that also hold a record in another country (Glänzel, 2001), and the association strength for a pair, the author-years with records in both countries over the number expected from their author-years (van Eck & Waltman, 2009). The counts agree with four of the conclusions that the maps reject, so on these the measure decides the conclusion while the records stay the same.

Table 1. Published conclusions against the maps, grouped by verdict. Each value is the change between the two windows in noise sd, signed so that positive values agree with the conclusion, and for a conclusion that ties held it is the change itself, which agrees inside the band. A level gives the rank of the country among the sixty-two countries by distance to the rest, nearest first, or the median rank of the pairs among the 780 pairs by direction similarity, closest first, and agrees on the side of the median the conclusion names. Equal and stable give the equal-volume and the stable-venue world, and counts read the records with the international share for countries and association strength for pairs.

Conclusion	Study	Measure in the study	Windows	Maps			Counts
				Main	Equal	Stable	
The maps disagree							
International coauthorship rises worldwide	Aksnes and Sivertsen (2023)	International coauthorship share	2008–2010, 2019–2021	−12.8	−13.4	−7.5	−44.9
The world of collaboration shrinks	Okamura (2023)	Copublication distance, clustering	2008–2010, 2018–2020	−9.5	−10.3	−5.3	−41.4
China joins the core	Gui et al. (2019a)	Coauthorship network analysis	2008–2010, 2013–2015	−5.8	−4.9	−24.2	+47.2
China stands outside the Euro-American core	Marginson (2022)	Review of science indicators	2008–2020	18th	24th	21st	61st
India stands semi-autonomous	Marginson (2022)	Review of science indicators	2008–2020	10th	10th	8th	51st
EU–Russia ties fall after the 2014 sanctions	Makkonen and Mitze (2023)	Gravity model of copublication	2011–2013, 2015–2017	−3.8	−1.4	−3.3	−10.6
EU–Russia ties hold after 2022	Hladchenko (2026)	Relative intensity of collaboration	2019–2021, 2023–2024	−4.2	−4.5	−4.1	−19.1
Russia turns from post-Soviet partners to the West	Matveeva et al. (2022)	Coauthorship shares by partner	2008–2010, 2016–2018	−4.0	−3.1	−2.4	+3.9
Ukraine turns towards Western partners	Matveeva et al. (2022)	Coauthorship shares by partner	2008–2010, 2016–2018	−2.1	−3.2	−4.1	−1.8

Table 1 (continued)

Conclusion	Study	Measure in the study	Windows	Maps			Counts
				Main	Equal	Stable	
Russia's international ties hold after 2022	Zhang et al. (2024)	Relative intensity of collaboration	2019–2021, 2023	+5.1	+1.8	+6.2	+10.4
US–Japan ties hold	Okamura (2023)	Jaccard distance of copublication	2008–2010, 2019–2021	−3.7	−3.6	−3.3	−23.7
Highly cited researchers move to Saudi affiliations	Catanzaro (2023)	Highly cited researchers' affiliations	2013–2015, 2021–2023	−3.8	−0.3	+1.8	−8.7
The maps agree							
Collaboration regionalises	Fitzgerald et al. (2021)	Community detection on coauthorship	2008–2010, 2016–2018	+5.3	+7.5	+1.9	+23.6
Colonial communities give way to regions	Fitzgerald et al. (2021)	Community detection on coauthorship	2008–2010, 2016–2018	+2.9	+1.9	+5.2	+10.0
Post-colonial pairs collaborate more	Gui et al. (2019a)	QAP regression on proximities	2008–2015	20th	18th	21st	28th
Russia loses centrality in Belt and Road science	Gui et al. (2019b)	Centrality and core–periphery	2008–2010, 2016–2018	+4.0	+3.1	+2.4	−3.9
The US coordinates the network	Gui et al. (2019a)	Coauthorship network analysis	2008–2015	1st	1st	1st	47th
China is central in Belt and Road science	Gui et al. (2019b)	Centrality and core–periphery	2008–2018	19th	25th	21st	61st
The US sits at the core of co-affiliation	Hottenrott et al. (2021)	Co-affiliation network	2008–2019	1st	1st	1st	44th
China sits at the core of co-affiliation	Hottenrott et al. (2021)	Co-affiliation network	2008–2019	18th	24th	21st	61st
Germany sits at the core of co-affiliation	Hottenrott et al. (2021)	Co-affiliation network	2008–2019	5th	6th	6th	39th
The UK sits at the core of co-affiliation	Hottenrott et al. (2021)	Co-affiliation network	2008–2019	2nd	3rd	3rd	23rd
South Korea stands semi-autonomous	Marginson (2022)	Review of science indicators	2008–2020	41st	47th	47th	52nd
Iran stands semi-autonomous	Marginson (2022)	Review of science indicators	2008–2020	57th	59th	59th	62nd
Iran is less connected than its peers	Kokabisaghi et al. (2019)	International coauthorship share	2010–2015	58th	59th	59th	62nd
The US sits at the centre of coauthorship	Leydesdorff et al. (2013)	Coauthorship network centrality	2011	1st	1st	1st	46th
Western EU countries form the core	Hladchenko (2026)	Relative intensity of collaboration	2008–2024	74th	97th	120th	124th
US–China ties converge until 2018	Kitajima and Okamura (2025)	Copublication distance	2008–2010, 2016–2018	+5.9	+7.9	+5.5	+36.6
US–China ties grow	Wagner et al. (2015a)	Coauthored paper counts	2008–2010, 2011–2013	+4.5	+4.4	+6.0	+19.4
US–China ties fall after the China Initiative	Geng et al. (2025)	Coauthorship networks	2015–2017, 2019–2021	+3.0	+2.5	+4.5	+9.2
US–China ties suffer from the NIH investigations	Jia et al. (2024)	Output of US scientists	2015–2017, 2019–2021	+3.0	+2.5	+4.5	+9.2

Table 1 (continued)

Conclusion	Study	Measure in the study	Windows	Maps			Counts
				Main	Equal	Stable	
US–China dual affiliations fall	Van Noorden (2022)	Authors with dual affiliations	2015–2017, 2019–2021	+3.0	+2.5	+4.5	+9.2
US–China ties diverge from 2019	Kitajima and Okamura (2025)	Copublication distance	2016–2018, 2020–2021	+4.5	+4.0	+4.8	+12.3
US–China ties weaken after 2019	Okamura (2023)	Jaccard distance of copublication	2016–2018, 2020–2021	+4.5	+4.0	+4.8	+12.3
China turns to domestic ties after 2018	Geng et al. (2025)	Coauthorship networks	2015–2017, 2019–2021	+4.5	+1.7	+9.9	−23.3
China's top research turns inward	Owens (2023)	Share-to-count ratio, Nature Index	2015–2017, 2020–2022	+10.6	+7.6	+15.9	−17.1
Russia turns towards the EU	Matveeva et al. (2022)	Bilateral coauthorship shares	2008–2010, 2016–2018	+3.0	+3.1	+2.9	+3.1
EU–Russia ties rise after 2014	Hladchenko (2026)	Relative intensity of collaboration	2011–2013, 2015–2017	+3.8	+1.4	+3.3	+10.6
Czechia and Poland sit close to Russia	Hladchenko (2026)	Relative intensity of collaboration	2008–2024	68th	71st	86th	98th
EU–Latin America ties grow	Belli and Baltà (2019)	Copublication counts	2008–2010, 2014–2016	+2.9	+0.9	+0.8	+2.3
Swiss ties with the EU hold after 2014	Saric et al. (2021)	Grant records and surveys	2011–2013, 2015–2017	−0.3	+1.1	+1.5	+7.3
UK–EU ties hold after Brexit	Hladchenko (2026)	Relative intensity of collaboration	2013–2015, 2017–2019	+1.4	+1.9	−0.4	−11.4
Saudi universities buy foreign affiliations	Bhattacharjee (2011)	Highly cited researchers' Saudi affiliations	2008–2010, 2012–2014	+10.7	+4.0	+2.6	+22.2
Foreign researchers drive Saudi science	Siddiqi et al. (2016)	Share of domestic corresponding authors	2008–2010, 2011–2013	+8.7	+2.4	+1.7	+18.8
Inside the band							
The global network keeps integrating	Gui et al. (2019a)	Coauthorship network analysis	2008–2010, 2013–2015	−0.4	−3.3	−0.7	−6.0
The global network grows denser	Wagner et al. (2015b)	Coauthorship network density	2008–2009, 2010–2011	−0.5	−2.2	−1.1	−0.9
International collaboration rises across fields	Coccia and Wang (2016)	International share by field	2008–2009, 2011–2012	−0.6	−3.8	−0.6	−3.6
US–China ties partly decouple	Marginson (2022)	Review of science indicators	2015–2017, 2019–2020	+2.0	+1.4	+2.8	+2.5
China–Russia ties rise	Haupt et al. (2023)	Copublication counts and shares	2013–2015, 2020–2022	+0.2	−0.9	+0.3	−2.8
Belarus turns towards Western partners	Matveeva et al. (2022)	Coauthorship shares by partner	2008–2010, 2016–2018	−0.6	+0.6	−1.5	+8.7
Ukrainian researchers gain foreign affiliations	Ganguli and Waldinger (2024)	Affiliations of Ukrainian scientists	2021, 2023	+1.2	+1.1	+4.4	+3.1
Swiss ties with the EU fall after 2014	Cavallaro and Lepori (2021)	Horizon 2020 participation	2011–2013, 2015–2017	+0.3	−1.1	−1.5	−7.3
UK–EU ties fall after the Brexit vote	Cavallaro and Lepori (2021)	Horizon 2020 participation	2013–2015, 2017–2019	−1.4	−1.9	+0.4	+11.4
UK–EU ties weaken after Brexit	Oldac and Olivos (2026)	Share of UK coauthored papers by region	2013–2015, 2017–2019	−1.4	−1.9	+0.4	+11.4
UK–China ties strengthen after Brexit	Oldac and Olivos (2026)	Share of UK coauthored papers	2013–2015, 2017–2019	−0.8	+1.0	+1.4	+8.7

Table 1 (continued)

Conclusion	Study	Measure in the study	Windows	Maps			Counts
				Main	Equal	Stable	
India–South Korea ties intensify	Dua et al. (2023)	Relative intensity of collaboration	2008–2010, 2018–2020	+1.6	+2.2	+4.0	−6.4
Hong Kong–Mainland ties grow	Ma and Li (2018)	Copublication counts and shares	2008–2010, 2015–2017	+1.3	−0.5	−1.2	−3.8
India internationalises	Dua et al. (2023)	International coauthorship share	2008–2010, 2018–2020	−0.7	−1.0	+0.4	+4.6
Iran collaborates more under sanctions	Kokabisaghi et al. (2019)	International share of Iranian papers	2010–2012, 2013–2015	−0.6	−0.6	−1.9	+2.1

9 Glossary

Record One paper, one author and one institution.

Author-year All records of one author in one year.

Half One of two random groups of authors, fixed across years.

Cell One year of one half, with its own map.

Weight An institution's fixed share of all author-years over all years, with each author-year split over its institutions in proportion to its records.

Centre of the world The weighted average of all prepared arrows on a map.

Centroid The weighted average position of a country's institutions.

Spread The typical distance of an institution from the centre.

Distance to the rest The gap between a country's centroid and the centroid of all other institutions, in units of the spread.

Direction similarity The cosine of the angle between two countries' arrows.

Main institution The institution with most of an author-year's records.

Twin world The same data with the years shuffled within each half and main institution.

Net value The real value minus the twin's value.

Noise sd The typical size of chance variation, from half the difference between the halves.

Noise band Two noise sd on either side of a line.

Net movement A country's net value minus its average net value over the years.

Common drift The average net movement of all countries.

Own movement A country's net movement minus the common drift, in noise sd.

Movement score The sum over the years of a pair's squared, centred net values in noise sd.

Field view The measures read inside one OpenAlex field, with each institution measured through its records in that field.

Lean A part's direction similarity with a neighbour minus that of the rest of its country.

International share The share of a country's co-affiliated author-years that also hold a record in another country.

Association strength The author-years with records in two countries divided by the number expected from the two countries' author-years.

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Supplementary material

S1 Decisions and their evidence

This chapter follows the pipeline from papers to figures and records each choice with its reason and, where a test informed it, the evidence. The tests ran with earlier versions of the code, mostly on the maps of 2017, and values that describe the current runs say so. In those earlier versions both halves started from the same random positions, which shares part of the training noise between them and raises every agreement between halves. The chapters after this one give the counts and settings of the current runs.

S1.1 Data

One release. Every world reads the OpenAlex release of 2026-09-23 (Priem et al., 2022) through one shared extraction, so all worlds rest on the same papers.

Six types of paper. Articles, reviews, letters, preprints and book chapters follow the document types of Murray et al. (2023). Conference papers join them as a type of their own in OpenAlex, since computing and several other fields publish much of their research at conferences.

2008 to 2025. The span opens with the first year studied by Murray et al. (2023) and closes with the last complete year of the release.

One map per calendar year. A yearly map ties institutions through one person in one year, by two affiliations on one paper, by a move within the year or by affiliations on different papers. A move from one year to the next puts its two institutions in two different maps, so each map holds the ties of its own year.

Papers with a primary topic. A paper enters when OpenAlex assigns it a primary topic. In the current selection this rule sets aside 84,753 papers, 0.11% of the papers of the six types in these years.

Every affiliation. Papers keep all their affiliations, however long their author lists. Each author-year carries a weight of one, so a paper with thousands of authors adds one record to each of their author-years, whose weights stay at one. A cap of 50 affiliations per paper in an earlier build removed 162,425 papers, 0.22% of the selection, and 27.0 million records.

Records. Each author of a paper gives one record for each distinct institution listed with them. Authorships enter when they carry an OpenAlex author identifier and at least one institution.

S1.2 Author-years and pairs

Two institutions to enter. An author-year enters when its records name at least two different institutions, since a tie needs two ends, and then all its records go in. Such author-years belong to the researchers who tie institutions together, the mobile researchers of Murray et al. (2023).

Repeated records stay. An author-year with four papers at one institution, one of which also names a second institution, keeps all five records, so the first institution is drawn four times as often as the second. Pairs of two records at one institution then become possible.

Halves fixed by author. Each author falls into one of two halves by the lowest bit of the SplitMix64 hash of their OpenAlex identifier and stays in that half in every year, so each half is a separate population in every year. A paper with authors in both halves feeds both, through the records of each author. A check world, the paper split, sends each paper to one half by the same hash of the paper's identifier, and its cells hold 1.23 times the author-years, 0.92 times the records and 0.61 times the papers of the main world (Table S1). The author-year is the unit of evidence, so the author split keeps every author-year whole and each researcher in one half, while the paper split cuts one person's year across both halves, which is why its cells hold more and thinner author-years. Its pooled noise sd is 1.05 times that of the main world for countries and 1.00 times for pairs, so splitting by paper gives noise of the same size. Its drawn series correlate with those of the main world at 0.99 for countries and 0.96 for pairs, and the own movement of countries at 0.81.

One weight per author-year. The sentence scheme of Murray et al. (2023) gives an author-year one pair for every neighbouring pair of its records, so its weight grows with its number of records. Author-years naming eight or more institutions hold at least eight records each. Under that scheme their share of the pairs rose from 10.4% in 2008 to

Table S1. The main world, with halves fixed by author, and the paper split, with halves fixed by paper, over the units both measure. Noise sd is the median over units of each unit's root mean square sd over the years, and the ratio divides the root mean squares pooled over all units, paper split over main world. Beyond two sd is the share of unit-years whose own movement, for pairs the net movement in noise sd, lies beyond two noise sd. Half agreement and the last two columns are correlations over all unit-years.

		Noise sd			Beyond two sd (%)		Half agreement		Correlation	
Measure	Units	Author	Paper	Ratio	Author	Paper	Author	Paper	Drawn	Own
Distance to the rest	62	0.00605	0.00620	1.05	31.6	28.3	0.598	0.591	0.989	0.815
Direction similarity	780	0.03402	0.03356	1.00	12.6	11.2	0.209	0.198	0.959	0.464

21.5% in 2025 (Figure S2). Over the same years their share of author-years rose from 2.1% to 5.0%. A gain of 11.1 percentage points against 2.9 would enter the maps as a trend. Every sweep over the data therefore draws exactly one pair from each author-year, so each person counts once in each year (Figure S1) and these author-years hold the same share of pairs as of author-years. In the earlier test on the maps of 2017, held-out prediction by the cosine with four pairs per author-year stayed close to that of the sentence scheme, at 0.842 against 0.852 overall. Across borders the two reached 0.338 and 0.337.



Figure S1. Pairs from two author-years. Ana's two records hold one pair and Ben's forty hold 780, since n records hold $n(n-1)/2$ pairs. Each author-year gives 20 training pairs.

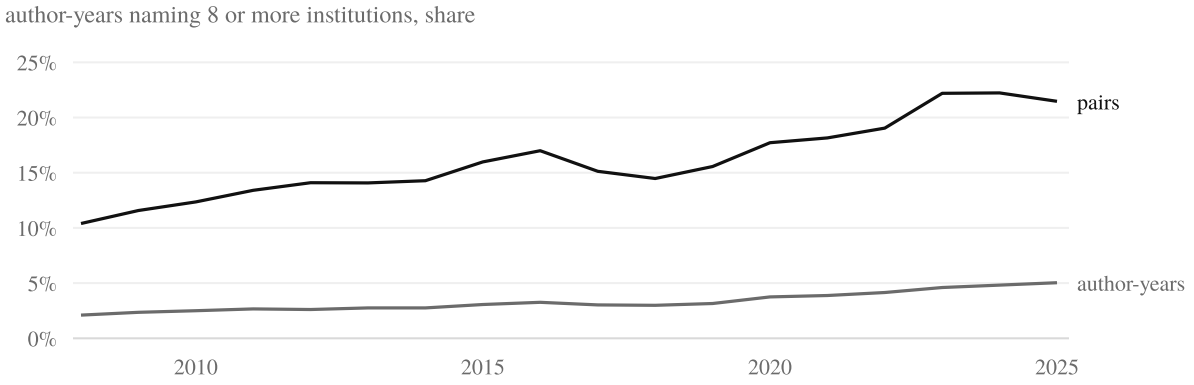


Figure S2. Author-years naming eight or more institutions, as a share of all author-years and as a share of the pairs they give under the sentence scheme, by year, with the two halves of the main world pooled. Each author-year gives one pair less than its number of records.

Two records per pair, same institution included. A pair takes its first record at random among the author-year's records and its second at random among the others, so two records at one institution can form a pair. These pairs pull each institution's input and output vectors towards one direction, and the cosine of input vectors, which the measures read, then carries the direct ties between institutions as well as their shared partners. With eight pairs per author-year in each pass, the two vectors of an institution reach a cosine of 0.31, against 0.19 when training skips these pairs. Skipping them lowers held-out prediction by the cosine to 0.778, against 0.847 with them, while the training target predicts equally well either way.

Four pairs per author-year in every pass. Each pass over the data makes four sweeps and training makes five passes, so every author-year gives 20 pairs to each map. The number of pairs follows a fixed rule (Table S2). The rule keeps the settings within 0.01 of the best held-out prediction both overall and across borders, then those within 0.01 of the best agreement between halves, and then takes the cheapest. Four pairs are the only setting to pass the first step, and eight predict 0.015 worse across borders and agree less between halves.

A new order for every sweep. Every sweep visits all author-years once, in a new order drawn from the model's seed, and takes one pair from each. The learning rate falls during training, so later pairs move the map less, and a new order in every sweep spreads each author-year's pairs over the whole run.

S1.3 Training

One model per cell, trained from scratch. Every cell and seed trains its own skip-gram model (Mikolov et al., 2013) from a random start, as in Murray et al. (2023), so each map rests on its own year alone. The measures read each map within its year, where every number stays the same under any rotation of the map.

Negatives from the exact marginal. The negatives of a pair are drawn in proportion to how often each institution appears as a target. Each author-year spreads a weight of one over its records, and an institution's chance is the sum of its shares, raised to the power one. The baseline of the comparison is then each institution's own weight, the idea behind residual2vec (Kojaku et al., 2021). With this baseline the training target at its optimum equals, up to a constant, the logarithm of the observed co-occurrence over the co-occurrence expected from the sizes of the two institutions, which links the maps to gravity models (Murray et al., 2023). The power of 0.75 used by word2vec would raise the weight of small institutions and shift that ratio.

128 dimensions. Under the same rule (Table S2), every setting from 128 to 512 dimensions ties on prediction and on agreement between halves, and 128 is the cheapest.

16 negatives. The same rule favours fewer negatives, since every setting up to eight lies within 0.01 of the best prediction and 16 falls 0.016 short across borders (Table S2). The study reads change across years, so the number of negatives follows how closely the two halves agree on change, a test that needs a full world for each setting (Table S3). The setting of 16 negatives leads on pairs and that of 32 on countries, and 16 costs less.

Table S2. Tests behind three settings on the maps of 2017, made with an earlier version of the code. Each block varies one setting around a common base of four pairs per author-year, 300 dimensions and five negatives. Held-out prediction fits the co-occurrence counted in one half with institution effects and the cosine of input vectors from the other half, and reports the share of the remaining deviance that the cosine explains, over all pairs of the 1,500 largest institutions and over those across borders. Half agreement correlates the cosines of the two halves over the 3,000 largest institutions. Both halves started from the same random positions in that version, which raises the agreement.

		Deviance explained		
Setting	Value	Overall	Across borders	Half agreement
Pairs	1	0.774	0.242	0.941
	4	0.842	0.338	0.895
	8	0.847	0.323	0.837
	16	0.832	0.299	0.751
	32	0.805	0.276	0.643
Dimension	32	0.808	0.281	0.868
	64	0.831	0.312	0.887
	128	0.840	0.332	0.894
	256	0.841	0.337	0.899
	300	0.842	0.338	0.895
	512	0.840	0.338	0.893
Negatives	2	0.839	0.342	0.906
	4	0.840	0.341	0.896
	5	0.842	0.338	0.895
	8	0.844	0.333	0.898
	16	0.843	0.326	0.888
	32	0.840	0.318	0.873

Inherited settings. The initial learning rate of 0.025 and its linear fall to 0.0001 times that value follow word2vec, as do the five passes and the clipping of scores at ±6. The sweeps above replaced the 300 dimensions and five negatives of Murray et al. (2023).

Two seeds per cell. Each cell trains twice, with seeds 11 and 12, and every value is averaged over the two maps. Both halves share their seeds and random streams, and each cell trains only the institutions present in it. An institution therefore starts from the same vector in both halves only when it holds the same place among the institutions of both cells, which applies to 0.04% to 3.93% of measured institutions over the years. In the current runs the training noises of the two halves correlate at −0.03 for countries and 0.00 for pairs.

Table S3. Agreement of the two halves on change, by number of negatives, in the main world and its twin trained with 128 dimensions and an earlier version of the code. Each value correlates the net changes of the two halves over 63 countries, for distance to the rest, or over 780 pairs, for the inner product of the two centroids that served as the pair measure in that version. Early changes run from 2008–2010 to 2017–2019 and late changes from 2017–2019 to 2023–2025. Both halves started from the same random positions in that version, which raises the agreement.

	Countries		Pairs	
Negatives	Early	Late	Early	Late
2	0.827	0.680	0.502	0.275
4	0.866	0.732	0.570	0.328
8	0.832	0.705	0.631	0.359
16	0.835	0.765	0.661	0.440
32	0.869	0.799	0.603	0.406

Reproducible runs. Every job trains one cell and seed on a single thread from seeded random streams, so a seed fixes its model whatever the number of jobs that run at once. The input vectors are stored in half precision, and rounding three maps of an earlier single-precision run to half precision moved distance to the rest by at most 0.00001 and direction similarity by at most 0.00004, under a hundredth of the smallest noise sd.

S1.4 Reading the maps

Input vectors and their cosine. Closeness between institutions is the cosine of their input vectors, as in Murray et al. (2023), so the maps and the numbers come from the same vectors.

Scale, centre and scale again. Raw vector length grows with how often an institution appears, so a centre removed from raw vectors mixes frequency into direction. The order of unit length, weighted centre and unit length was chosen by how well cosines from one half predict the excess co-occurrence counted in the other. In the current runs, averaged over the years and with each half predicting the other, this order explains 0.69 among the 3,000 institutions of largest weight and 0.16 among the 3,000 of smallest weight. Raw vectors centred by weight or by records explain 0.68 and 0.67 among the largest and 0.09 and 0.08 among the smallest. Unit length alone explains 0.62 and 0.14.

The same institutions every year. The measures read a fixed set of 19,254 institutions, those with a known country and at least three records in every cell of the main world and of its twin. A fixed set keeps the entry and exit of institutions out of the measured movement. Higher floors of up to 16 records shrink the set to as few as 10,287 institutions in 49 countries with a distance series (Table S4). Own movement then correlates at 0.96 or more with that at the floor of three.

Table S4. The main world measured with floors of 3, 4, 8 and 16 records in every cell. Countries counts the countries with a distance series, and half agreement correlates the centred net values of the two halves over all unit-years. The last four columns correlate each floor's values with those at the floor of three over the units both measure, for the mean distance to the rest and own movement of each country and the mean direction similarity and movement score of each pair.

			Half agreement		Correlation with floor 3			
Floor	Institutions	Countries	Distance	Direction	Level	Own	Pair level	Score
3	19,254	62	0.598	0.209				
4	17,574	60	0.587	0.209	0.999	0.993	1.000	0.998
8	13,723	56	0.611	0.203	0.996	0.978	0.999	0.994
16	10,287	49	0.603	0.169	0.995	0.959	0.997	0.983

Weights from author-year shares over the whole period. Training counts every author-year once, so the measures weigh institutions by their shares of author-years, and each institution keeps its share over all 36 cells in every year.

Institutions present in the cell. In the main world and its twin every institution of the set passes the floor in every cell. A check world measures the institutions that pass its own floor in every cell, and its twin measures those of them that appear in each of its cells. The weights are renormalised over the institutions measured, and Table S10 gives the share left out.

Countries as OpenAlex codes them. Each institution keeps the country code that OpenAlex gives it in every year, with Hong Kong, Macau and Taiwan as separate codes.

S1.5 Measures

Measures inside one year. Every number is a distance or an angle inside one year's map, and only the numbers travel across years. Earlier versions described each institution by its cosines to a set of anchor institutions, and an anchor set re-weights the directions of the map. Relative to equal weighting, the 512 most active anchors weighted the first direction by 19, the second by 10 and the third by 5. Random sets weighted the first direction by up to 31, the results moved with the choice of set, and the measures therefore read the map directly.

Distance to the rest. A country's distance to the rest is the distance between the weighted centroid of its institutions and that of all other institutions, divided by the spread $\sqrt{S}$ of the year's map, which gives every year the same unit. The measure is a function of the cosines within the year, and the squared distance between the two centroids equals the average closeness inside the country plus that inside the rest minus twice that between the two (Section S10).

Reporting thresholds. A country gets a distance series when the set holds at least 20 of its institutions, which gives 62 countries, and the pairs join the 40 countries with the largest total weight, which gives 780 pairs. Both thresholds are reporting choices, since countries with few institutions have wide bands, and a country with few but large institutions can join the pairs below the threshold for a distance series.

Direction similarity. Two countries are compared by the cosine of the angle between their arrows from the centre of the world. The lengths of the arrows already appear as distances to the rest, so the angle carries what the lengths leave open, and together the two measures hold the full geometry of the country centroids.

S1.6 Comparing years

A shuffled twin for every world. Years with more data spread a map over more dimensions and leave a smaller share to the contrasts between countries, so raw values drift with the volume of data. The twin moves whole author-years between years within each half and each main institution, the institution an author-year names most often, with ties broken at random. Every cell keeps its number of author-years and every main institution its count in each year, while the timing of the ties is mixed, so the twin follows the drift while its structure stays that of the average year. Strata by institution keep growth that concentrates inside one country, and a twin stratified by country gives a drift that correlates 0.92 with this one.

Net values, half by half. Each value is first averaged over the two seeds, the twin's value is then subtracted within each half, and a series averages the two halves. Seeds add only training noise inside a half, and keeping the halves apart to the end leaves two independent estimates for the noise. The equal-volume check tests the subtraction (Table S17).

Noise from the halves. Half the difference between the two halves' net values is a series of noise alone, since a true change appears in both halves and cancels. The two noises are independent, so the half-difference has the same size as the noise in their average. In the current runs the draw of people into halves makes up 74% of this noise for countries and 26% for pairs, and training makes up the rest, so more seeds would mostly narrow the bands of pairs.

One noise profile over the years. The noise sd is one scale per unit times one profile over the years, shared by all countries or by all pairs, $\mathrm{sd}_{u,t} = s_u P_t$. Each band then rests on the unit's whole series, and the profile follows the larger noise of the early years, when cells are small. Table S13 reports how well the model fits.

Bands of two sd. The bands span two noise sd on either side, a convention under which about 4.6% of noise values fall outside.

Whole trajectories. Every figure shows whole series against their bands. Tests for a break would add assumptions about shape, such as a single break, straight segments and constant noise, while a series can turn more than once.

S1.7 Display

Level from the real world, movement from the net series. A series is drawn at the height of its real mean over the 18 years, with the year-to-year shape of its net value. The net value alone sits near zero, since subtracting the twin also removes the average structure of the map.

Common drift and own movement. The common drift is the unweighted mean of the 62 countries' centred net distances, so every country counts once, and its band treats their noises as independent. Own movement subtracts the drift from a country's centred net distance and divides by the country's noise sd, which puts every country on one scale.

A movement score for pairs. A pair's score sums its squared centred values over the years, each in units of its noise sd, so noise alone gives about 17. The score treats every shape of change alike, and pairs keep their share of the common change.

Orders from the data. Rows of the movement map follow the first principal pattern of own movement, the eight highlighted countries have the largest sums of squared own movement, and the similarity matrix follows an average-linkage clustering of the countries. These rules choose every year and country that the figures single out.

S1.8 Checks

Two checks, each with its own twin. Each check is a full rerun with its own twin. It is compared with the main world over the units they share by correlating their year-to-year shapes, the values minus each unit's mean, over all unit-years, for the net and for the real values.

Equal volume. Every cell of this world holds 603,599 author-years, the size of the smallest main cell, drawn at random as whole author-years. Its training volume stays the same in every year, so its net changes should agree with those of the main world when the twin removes the volume drift. Read without its own twin, it agrees with the main net values at 0.48 for distance to the rest, against 0.72 with it, since at a fixed total the growth still differs between institutions, and for direction similarity the two reach 0.44 and 0.41. It stays a check because it leaves out about two thirds of each 2025 cell.

Stable venues. This world keeps the papers of the 15,717 venues with selected papers in every year from 2008 to 2025. OpenAlex coverage changes as sources enter and leave, which alters the content of the data at similar volume, and the twin holds only volume fixed. A paper needs a venue to enter this world, which therefore holds a smaller part of the conference papers.

S2 Data

All counts come from the OpenAlex release of 2026-09-23, restricted to papers published from 2008 to 2025 as articles, book chapters, conference papers, letters, preprints and reviews, each with a primary topic in OpenAlex. Table S5 follows the selection from papers to the author-years that enter the maps, Table S6 gives the same counts for every year, and Table S7 lists the 36 cells of the main world.

Table S5. From papers to the author-years that enter the maps.

Quantity	Value
Papers selected	74,293,950
Records of selected papers	399,416,530
Author-years entering	41,379,034
Records entering	276,256,727
Institutions in the vocabulary	119,028
Cells	36
Years	2008–2025
OpenAlex release	2026-09-23

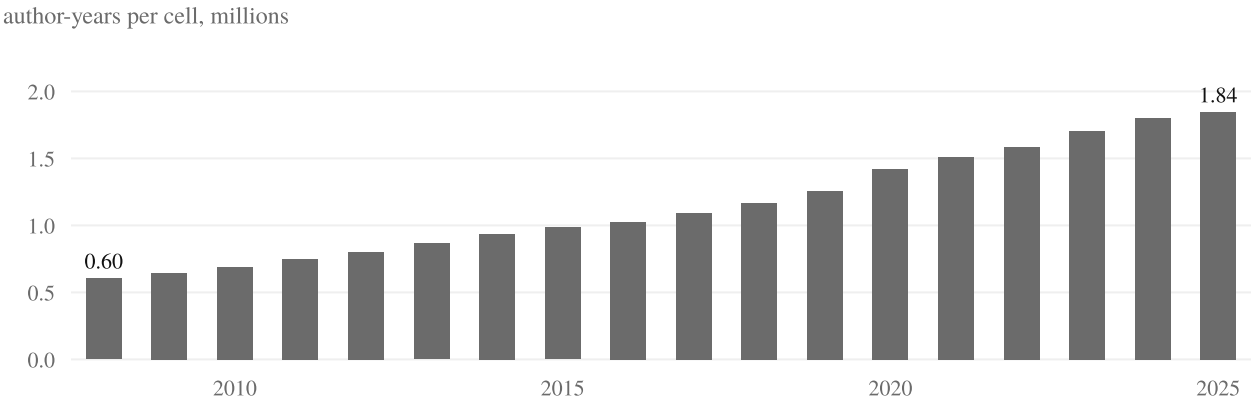


Figure S3. Author-years per cell in millions, as the mean of the two halves.

A paper with authors in both halves appears in both, so the papers summed over the cells exceed the papers selected.

Table S6. Papers, records and author-years by year.

Year	Papers	Records	Author-years entering	Records entering
2008	2,505,912	10,079,908	1,207,482	6,628,300
2009	2,693,052	10,988,372	1,288,408	7,247,036
2010	2,859,421	11,985,043	1,381,964	7,930,540
2011	3,083,197	13,437,616	1,498,767	8,939,707
2012	3,206,503	14,793,700	1,604,849	9,946,890
2013	3,382,755	15,911,024	1,730,829	10,725,476
2014	3,539,720	17,304,605	1,877,463	11,799,840
2015	3,601,143	18,378,288	1,969,177	12,694,363
2016	3,681,088	19,722,059	2,050,308	13,686,933
2017	3,909,362	21,046,977	2,181,061	14,477,095
2018	4,172,294	22,795,180	2,331,889	15,648,854
2019	4,488,632	24,940,394	2,519,005	17,204,120
2020	5,014,817	29,070,086	2,845,500	20,540,324
2021	5,271,300	31,158,593	3,023,346	21,915,069
2022	5,319,893	32,099,935	3,169,453	22,372,909
2023	5,620,580	34,386,192	3,409,876	24,474,533
2024	5,869,340	35,639,062	3,602,103	25,142,533
2025	6,074,941	35,679,496	3,687,554	24,882,205

Table S7. The 36 cells of the main world.

Year	Half	Papers	Author-years	Records
2008	a	1,185,633	603,599	3,307,667
	b	1,197,758	603,883	3,320,633
2009	a	1,278,033	644,558	3,625,059
	b	1,277,412	643,850	3,621,977
2010	a	1,368,995	691,200	3,967,914
	b	1,376,191	690,764	3,962,626
2011	a	1,483,667	748,842	4,469,563
	b	1,488,385	749,925	4,470,144
2012	a	1,558,872	801,924	4,971,876
	b	1,563,365	802,925	4,975,014
2013	a	1,681,015	865,366	5,370,014
	b	1,678,571	865,463	5,355,462
2014	a	1,807,943	938,215	5,908,355
	b	1,806,088	939,248	5,891,485
2015	a	1,856,915	983,751	6,350,602
	b	1,860,416	985,426	6,343,761
2016	a	1,931,309	1,025,450	6,851,132
	b	1,932,729	1,024,858	6,835,801
2017	a	2,046,038	1,089,882	7,241,505
	b	2,050,552	1,091,179	7,235,590
2018	a	2,203,456	1,165,386	7,818,728
	b	2,203,646	1,166,503	7,830,126
2019	a	2,405,167	1,259,303	8,606,223
	b	2,403,744	1,259,702	8,597,897
2020	a	2,776,561	1,421,465	10,272,132
	b	2,778,815	1,424,035	10,268,192
2021	a	2,948,512	1,511,125	10,955,068
	b	2,947,941	1,512,221	10,960,001
2022	a	3,008,292	1,583,343	11,181,941
	b	3,012,419	1,586,110	11,190,968
2023	a	3,222,642	1,706,239	12,243,020
	b	3,220,666	1,703,637	12,231,513
2024	a	3,376,845	1,800,583	12,581,676
	b	3,378,294	1,801,520	12,560,857
2025	a	3,447,092	1,843,690	12,440,057
	b	3,447,086	1,843,864	12,442,148

S3 Training

Every cell trains skip-gram with negative sampling (Mikolov et al., 2013) on its pairs. In each pair the first record is the context c and the second the target t. Every institution has an input vector v and an output vector u, and each pair maximises

$$\log \sigma(u_t^\top v_c) + \sum_{k=1}^{K} \log \sigma(-u_{n_k}^\top v_c),$$

where σ is the logistic function and the K negatives n_k are drawn from the marginal of the targets. Each author-year spreads a weight of one over its records, and an institution's chance of being drawn is the sum of its shares. The measures read the input vectors. Table S8 lists every setting, and Section S1 gives the reasons for them.

Table S8. Training and measure settings.

Setting	Value
Dimension	128
Negatives per pair	16
Noise exponent	1
Initial learning rate	0.025
Final learning rate, share of initial	0.0001
Passes over the data	5
Pairs per author-year and pass	4
Pairs per author-year and map	20
Seeds	11, 12
Maps per world	72
Worlds, each with its own twin	3
Minimum records in every cell, measured set	3
Minimum institutions for a distance series	20
Countries in the pairs	40

S4 Institutions and weights

The measures read the 19,254 institutions that have a known country and at least three records in every cell of the main world and of its twin. Institutions carry the weights of Section 4, and each map renormalises them over the institutions present in its cell. Table S9 gives the institutions and weight of every country in the two measures. The full list of institutions is available as a file on the methods site.

Table S9. Measured institutions and weight for the 63 countries in either measure, ordered by weight. The 40 countries above the rule form the pairs, and Hong Kong enters the pairs only.

Country	Institutions	Weight (%)
China	2,126	21.79
United States	3,592	18.78
France	1,743	6.87
United Kingdom	997	4.30
Germany	877	4.18
Japan	1,166	3.66
Brazil	501	2.82
Italy	566	2.79
India	648	2.49
Spain	581	2.37
Canada	438	2.33
South Korea	362	2.27
Australia	368	2.03
Russia	448	1.76
Netherlands	222	1.53
Taiwan	215	1.05
Switzerland	210	1.03
Poland	197	0.89
Iran	174	0.88
Türkiye	179	0.86
Sweden	121	0.73
Argentina	167	0.69
Portugal	135	0.68
Belgium	117	0.67
Mexico	168	0.66
Indonesia	49	0.59
Denmark	89	0.57
Czechia	119	0.49
Austria	125	0.45
Malaysia	60	0.43
Finland	73	0.43
Israel	83	0.41
Singapore	68	0.40
Greece	108	0.40
Hong Kong	17	0.38
Norway	91	0.37
Pakistan	78	0.36
Egypt	63	0.34
South Africa	73	0.32
Tunisia	48	0.31
Saudi Arabia	42	0.31
Thailand	61	0.28
Chile	83	0.28
Ukraine	79	0.26
Nigeria	67	0.26
Ireland	68	0.25
Hungary	76	0.23
Romania	64	0.21
New Zealand	57	0.20
Colombia	71	0.19
Serbia	49	0.19
Vietnam	33	0.17
Morocco	32	0.17
Slovakia	64	0.16
Bangladesh	38	0.14
Croatia	40	0.11
Bulgaria	44	0.11
Algeria	37	0.10
Slovenia	30	0.09
Kenya	25	0.06
Peru	25	0.06
Belarus	21	0.05
Cuba	30	0.05

Table S10. Measured institutions absent from a cell, by world.

World	Largest number absent from a cell	Weight (%)
Main	0	0.000
Main twin	0	0.000
Equal volume	3,304	1.956
Equal-volume twin	3,310	1.959
Stable venues	3,997	4.382
Stable-venue twin	4,001	4.385
Paper split	1,832	1.688
Paper-split twin	1,834	1.689

S5 The shuffled twin

The twin shuffles the years of the author-years within each half and each main institution, with its own random stream for each half. Table S11 reports, for every world and half, the author-years shuffled, the number of main institutions that form the groups and the share of author-years whose year changed. Records at secondary institutions travel with their author-years, so the yearly count of a secondary institution holds only approximately.

Table S11. The shuffled twins, by world and half.

World	Half	Author-years shuffled	Main institutions	Year changed (%)
Main	a	20,683,921	92,371	92.4
	b	20,695,113	92,470	92.4
Equal volume	a	10,864,782	83,960	92.7
	b	10,864,782	84,079	92.7
Stable venues	a	13,883,503	81,879	92.6
	b	13,889,665	81,729	92.6
Paper split	a	25,480,710	93,487	92.2
	b	25,488,571	93,517	92.3

S6 Noise

Every value is first averaged over the two seeds. For each half, the net value is the real value minus the twin's, and a series averages the net values of the two halves. The noise series is half the difference between the halves. Its standard deviation is modelled as

$$\mathrm{sd}_{u,t} = s_u \, P_t,$$

with one scale s_u per unit and one profile P_t over years shared by all units of a kind, the 62 countries or the 780 pairs. Table S12 gives the two profiles. If the model fits, about 4.6% of the noise values fall outside two sd, and Table S13 reports the observed shares.

The common drift δ_t is the unweighted mean of the centred net distances $\tilde{N}_{c,t}$ over the 62 countries, and its band treats the countries' noises as independent. Own movement is

$$o_{c,t} = \frac{\tilde{N}_{c,t} - \delta_t}{\mathrm{sd}_{c,t}}.$$

Table S12. Noise profiles over the years, for countries and for pairs.

Year	Countries	Pairs
2008	1.63	1.14
2009	1.17	1.16
2010	1.40	1.08
2011	1.25	1.10
2012	1.26	1.10
2013	1.23	0.99
2014	1.01	1.00
2015	0.91	1.02
2016	0.98	1.07
2017	0.97	0.97
2018	0.85	0.97
2019	0.92	0.96
2020	0.76	0.95
2021	0.79	0.93
2022	0.81	0.91
2023	0.67	0.88
2024	0.66	0.87
2025	0.75	0.89

Table S13. Calibration of the noise model. Under the model, 4.6% fall outside two sd.

Units	Outside two sd (%)	Lag-one autocorrelation
Countries	5.0	−0.03
Pairs	4.2	0.00

S7 Full results

Table S14 gives each country's mean distance to the rest, its typical noise sd, its drawn values in the first and last years and the extremes of its own movement with their years. Table S15 gives the common drift by year. Table S16 lists the 40 pairs with the largest movement scores, and the full table of 780 pairs is available as a file on the methods site.

Table S14. Distance to the rest by country. The year columns give the drawn series, the real mean plus the centred net value, and the noise sd is its mean over the years.

Country	Mean	Noise sd	2008	2025	Largest own (year)	Smallest own (year)
Algeria	0.870	0.0080	0.874	0.873	+2.9 (2017)	−3.0 (2023)
Argentina	0.891	0.0039	0.873	0.898	+3.1 (2015)	−3.4 (2010)
Australia	0.631	0.0040	0.634	0.644	+3.2 (2023)	−2.6 (2013)
Austria	0.688	0.0073	0.705	0.701	+5.0 (2009)	−3.4 (2017)
Bangladesh	0.611	0.0120	0.626	0.615	+1.8 (2009)	−2.3 (2021)
Belarus	0.720	0.0120	0.709	0.721	+3.6 (2024)	−2.8 (2018)
Belgium	0.673	0.0055	0.667	0.676	+1.6 (2018)	−3.4 (2019)
Brazil	0.770	0.0020	0.764	0.766	+3.7 (2023)	−4.9 (2025)
Bulgaria	0.771	0.0092	0.717	0.792	+6.4 (2022)	−3.1 (2008)
Canada	0.582	0.0049	0.582	0.583	+1.7 (2016)	−1.3 (2012)
Chile	0.815	0.0073	0.810	0.807	+1.6 (2009)	−2.1 (2025)
China	0.712	0.0013	0.686	0.743	+28.1 (2025)	−8.8 (2008)
Colombia	0.816	0.0089	0.824	0.798	+2.6 (2009)	−4.0 (2022)
Croatia	0.856	0.0069	0.830	0.866	+1.8 (2009)	−2.0 (2024)
Cuba	0.818	0.0100	0.784	0.832	+3.8 (2019)	−1.7 (2023)
Czechia	0.745	0.0044	0.730	0.742	+3.4 (2018)	−4.8 (2012)
Denmark	0.764	0.0042	0.760	0.764	+2.2 (2014)	−3.6 (2023)
Egypt	0.769	0.0076	0.760	0.778	+2.7 (2012)	−2.5 (2014)
Finland	0.779	0.0057	0.797	0.786	+3.5 (2009)	−2.3 (2023)
France	0.721	0.0020	0.709	0.735	+6.9 (2025)	−3.6 (2011)
Germany	0.588	0.0018	0.589	0.591	+4.2 (2014)	−7.3 (2018)
Greece	0.800	0.0054	0.785	0.814	+3.8 (2024)	−3.4 (2009)
Hungary	0.817	0.0064	0.773	0.814	+3.6 (2016)	−4.3 (2009)

Table S14 (continued)

Country	Mean	Noise sd	2008	2025	Largest own (year)	Smallest own (year)
India	0.676	0.0046	0.653	0.689	+3.6 (2023)	−2.9 (2012)
Indonesia	0.755	0.0065	0.747	0.732	+4.0 (2013)	−5.8 (2024)
Iran	0.842	0.0036	0.857	0.833	+4.0 (2008)	−7.7 (2023)
Ireland	0.726	0.0076	0.714	0.726	+2.5 (2020)	−3.2 (2024)
Israel	0.787	0.0055	0.786	0.783	+2.1 (2012)	−1.7 (2025)
Italy	0.692	0.0027	0.687	0.698	+3.0 (2014)	−5.5 (2023)
Japan	0.737	0.0025	0.717	0.740	+3.2 (2014)	−3.2 (2008)
Kenya	0.708	0.0191	0.711	0.700	+1.6 (2011)	−2.3 (2021)
Malaysia	0.773	0.0069	0.810	0.774	+4.0 (2008)	−2.6 (2011)
Mexico	0.775	0.0057	0.742	0.790	+2.7 (2025)	−2.8 (2008)
Morocco	0.864	0.0079	0.820	0.878	+2.7 (2018)	−2.8 (2008)
Netherlands	0.673	0.0027	0.676	0.673	+3.4 (2011)	−4.2 (2022)
New Zealand	0.757	0.0049	0.745	0.780	+6.0 (2023)	−3.0 (2011)
Nigeria	0.779	0.0130	0.808	0.725	+3.6 (2010)	−6.0 (2024)
Norway	0.735	0.0072	0.744	0.735	+2.5 (2009)	−1.9 (2016)
Pakistan	0.674	0.0079	0.687	0.682	+4.7 (2010)	−5.2 (2015)
Peru	0.777	0.0097	0.753	0.768	+3.5 (2015)	−3.1 (2024)
Poland	0.758	0.0039	0.741	0.756	+3.3 (2013)	−4.3 (2024)
Portugal	0.787	0.0036	0.764	0.797	+3.1 (2020)	−2.6 (2008)
Romania	0.754	0.0105	0.772	0.782	+3.2 (2025)	−3.7 (2012)
Russia	0.761	0.0029	0.735	0.783	+8.4 (2025)	−4.0 (2008)
Saudi Arabia	0.756	0.0064	0.852	0.761	+9.8 (2008)	−6.0 (2017)
Serbia	0.865	0.0083	0.825	0.881	+4.4 (2020)	−3.5 (2009)
Singapore	0.765	0.0061	0.777	0.762	+2.0 (2017)	−2.3 (2024)
Slovakia	0.785	0.0060	0.770	0.772	+2.9 (2022)	−3.7 (2025)
Slovenia	0.768	0.0073	0.738	0.786	+5.7 (2024)	−2.9 (2015)
South Africa	0.756	0.0048	0.726	0.766	+4.3 (2022)	−3.7 (2010)
South Korea	0.772	0.0019	0.758	0.779	+3.8 (2016)	−3.1 (2022)
Spain	0.716	0.0022	0.711	0.714	+4.4 (2009)	−9.3 (2024)
Sweden	0.711	0.0053	0.720	0.710	+2.3 (2010)	−3.4 (2022)
Switzerland	0.618	0.0067	0.618	0.613	+2.2 (2022)	−2.1 (2016)
Taiwan	0.775	0.0031	0.763	0.769	+1.9 (2015)	−4.3 (2025)
Thailand	0.770	0.0068	0.774	0.758	+7.1 (2011)	−4.7 (2014)
Tunisia	0.769	0.0127	0.720	0.788	+3.9 (2024)	−2.3 (2009)
Türkiye	0.826	0.0041	0.795	0.825	+5.0 (2017)	−6.3 (2011)
Ukraine	0.837	0.0067	0.808	0.824	+3.2 (2020)	−3.4 (2025)
United Kingdom	0.546	0.0036	0.538	0.539	+2.2 (2024)	−4.1 (2025)
United States	0.427	0.0017	0.419	0.435	+12.1 (2023)	−6.2 (2011)
Vietnam	0.692	0.0150	0.724	0.717	+2.4 (2024)	−2.4 (2010)

Table S15. Common drift by year.

Year	Common drift	Noise sd
2008	−0.0073	0.0015
2009	−0.0049	0.0011
2010	−0.0053	0.0013
2011	−0.0056	0.0011
2012	−0.0051	0.0011
2013	−0.0061	0.0011
2014	−0.0086	0.0009
2015	−0.0018	0.0008
2016	−0.0016	0.0009
2017	−0.0014	0.0009
2018	−0.0002	0.0008
2019	+0.0026	0.0008
2020	+0.0047	0.0007
2021	+0.0083	0.0007
2022	+0.0101	0.0007
2023	+0.0089	0.0006
2024	+0.0101	0.0006
2025	+0.0034	0.0007

Table S16. The 40 pairs with the largest movement scores. The year columns give the drawn series, the real mean plus the centred net value.

Pair	Mean similarity	2008	2025	Score
The United States and India	−0.020	−0.192	0.156	674
China and Malaysia	−0.068	−0.060	0.108	562
China and Pakistan	−0.035	−0.033	0.007	493
China and the United States	−0.176	−0.218	−0.243	327
Australia and Malaysia	0.282	0.274	0.201	276
The United States and Iran	−0.101	−0.217	0.020	208
China and India	−0.316	−0.198	−0.442	198
Switzerland and Pakistan	−0.160	−0.090	−0.216	195
The United States and Pakistan	−0.353	−0.521	−0.221	190
Iran and Malaysia	0.284	0.431	0.128	182
China and Germany	−0.345	−0.386	−0.324	148
Türkiye and Tunisia	0.341	0.093	0.344	145
The United States and Indonesia	−0.447	−0.531	−0.413	144
The United States and Italy	−0.064	−0.021	−0.093	140
India and Pakistan	0.433	0.531	0.387	131
The United States and Egypt	−0.052	−0.206	0.029	129
France and Pakistan	−0.139	−0.035	−0.251	122
France and Tunisia	0.439	0.500	0.370	118
Germany and Pakistan	−0.077	−0.042	−0.154	115
Pakistan and South Africa	0.167	0.200	0.021	111
Russia and Sweden	0.020	0.041	−0.060	106
China and Indonesia	−0.198	−0.090	−0.196	101
Portugal and Czechia	0.135	0.041	0.136	96
Austria and Pakistan	0.025	0.119	−0.073	96
The United States and South Africa	−0.037	−0.230	0.037	95
Türkiye and Pakistan	0.277	0.220	0.297	95
Belgium and Austria	0.217	0.143	0.195	93
The United States and the United Kingdom	−0.032	−0.013	0.013	93
The Netherlands and Austria	0.223	0.197	0.225	92
Poland and Pakistan	0.179	0.230	0.246	91
Hong Kong and Pakistan	−0.077	−0.162	0.055	90
The United Kingdom and Brazil	0.002	−0.054	−0.003	90
The United Kingdom and Australia	0.281	0.318	0.251	89
Spain and Mexico	0.522	0.499	0.530	89
Spain and Türkiye	0.214	0.076	0.176	89
China and Hong Kong	0.498	0.474	0.544	88
Italy and Malaysia	0.019	0.101	−0.046	86
Russia and Tunisia	0.256	0.152	0.279	84
India and Indonesia	0.384	0.490	0.360	83
India and Malaysia	0.287	0.394	0.226	80

S8 Check worlds

The equal-volume world holds 21.7 million author-years, and the stable-venue world holds 27.8 million after 35.6 million papers leave with their venues. Each check is compared with the main world over the units they share, by correlating their centred values over all unit-years, and Table S17 reports these correlations for the net and the real values of both measures.

Table S17. Agreement of each check with the main world, as the correlation of year-to-year shapes over all unit-years the two worlds share, for net values and for real values. The third row reads each check without its twin, as its real values against the net values of the main world. A shape is a series minus the unit's mean.

Measure	Values	Equal volume	Stable venues
Distance to the rest	Net	0.72	0.63
	Real	0.72	0.67
	Without twin	0.48	0.35
Direction similarity	Net	0.41	0.38
	Real	0.68	0.68
	Without twin	0.44	0.41

S9 How the map viewer draws the maps

The interactive map at clarelab.github.io/scholar-mm draws the yearly maps in three dimensions. Its pictures serve display only, and every number in this document comes from the within-year measures. The viewer averages each year's vectors over halves and seeds after a weighted generalised Procrustes alignment of all maps over all institutions (Gower, 1975). A weighted principal component analysis pooled over all years gives the axes, and the twin rescales each year's spread along every axis, which takes the volume drift out of the pictures. In the 2017 maps of an earlier version, the first two axes held about a quarter of the structure that repeats across halves. Aligned UMAP (McInnes et al., 2018) kept about a third of each institution's ten nearest neighbours against 2% for the first two axes, so the viewer offers aligned UMAP as a second view. Table S18 lists the settings.

Table S18. Settings of the map viewer.

Setting	Value
Institutions shown	14,002
Minimum records in every cell	8
Principal components	5
Variance shares of the components (%)	14.8, 7.9, 6.4, 4.0, 3.6
Procrustes iterations	6
UMAP components	3
UMAP neighbours	15
UMAP minimum distance	0.1
UMAP metric	cosine
UMAP alignment window	2
Neighbours listed	10

S10 Axes and what the measures mean

S10.1 Principal axes

Each map has 128 dimensions. We read them with a weighted principal component analysis of every map of the main world, on the prepared positions $z_i = (x_i - m)/\sqrt{S}$ with the cohort weights w_i. The weighted variances of z sum to one, so every axis carries a share of the map. The axes of the four maps of a year are matched to those of its first map by the order and signs that maximise the correlation of institution scores, and the scores are averaged over seeds and then over halves. A pooled version aligns all 72 maps by a weighted generalised Procrustes fit (Gower, 1975) and runs one analysis over the years.

The first axis holds 11.1% of the variance and the first eight hold 35.7% (Table S19). Half agreement falls from 0.98 on the first axis to 0.76 on the last in the yearly analysis, and the pooled analysis keeps the last axis at 0.89 (Figure S4).

Table S19. The first eight axes, averaged over the years. Match is the mean absolute correlation of institution scores with the first map of the year in the yearly analysis, and with the closest yearly axis in the pooled analysis.

Analysis	Axis	Variance share (%)	Half agreement	Match
Yearly	1	11.1	0.98	0.98
	2	6.3	0.96	0.96
	3	4.7	0.95	0.94
	4	3.3	0.87	0.83
	5	2.9	0.79	0.74
	6	2.7	0.73	0.70
	7	2.6	0.75	0.72
	8	2.2	0.76	0.74
Pooled	1	14.7	0.99	1.00
	2	7.8	0.97	0.93
	3	6.3	0.96	0.88
	4	4.0	0.92	0.93
	5	3.6	0.92	0.82
	6	3.4	0.92	0.76
	7	3.0	0.90	0.84
	8	2.7	0.89	0.92

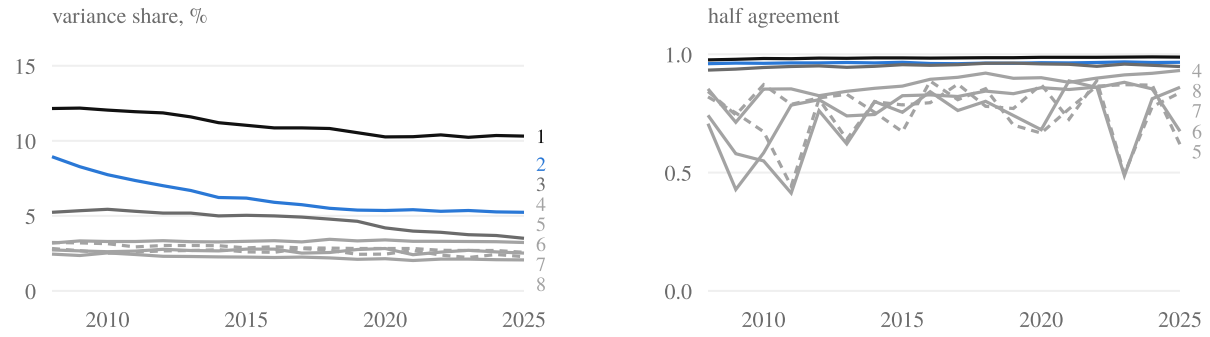


Figure S4. Variance share and half agreement of the first eight axes by year, in the yearly analysis.

S10.2 What the axes follow

Table S20 gives the weighted R^2 of each axis on six descriptions of the institutions. Partner countries are the countries of the other institutions in an institution's author-years, and fields are the shares of its records in each OpenAlex field, where a record takes the field of one work its paper cites, drawn at random among the cited works that have a topic. Both are counted in the year's cells. Type is the OpenAlex institution type, size is the logarithm of the institution's weight, and place is the position on the globe from latitude and longitude. On the first axis, partner countries explain the most, with $R^2 = 0.94$, and the institution's own country nearly as much, with $R^2 = 0.90$, so the first axis sorts institutions largely by country. Figure S5 names the institutions and countries at both ends of the leading axes.

S10.3 Distance to the rest in cosines

After the three moves every position has length one, so the inner product of two positions is their cosine. With $\bar{C}_{cd}$ the weighted mean cosine between the institutions of groups c and d, distance to the rest satisfies

$$D_c^2 S = \bar{C}_{cc} + \bar{C}_{\bar{c}\bar{c}} - 2\bar{C}_{c\bar{c}}, \qquad \bar{C}_{cd} = \sum_{i \in c} \sum_{j \in d} \frac{w_i w_j}{W_c W_d} \cos(x_i, x_j).$$

Distance to the rest grows with the mean cosine inside the country and inside the rest, and falls with the mean cosine between them.

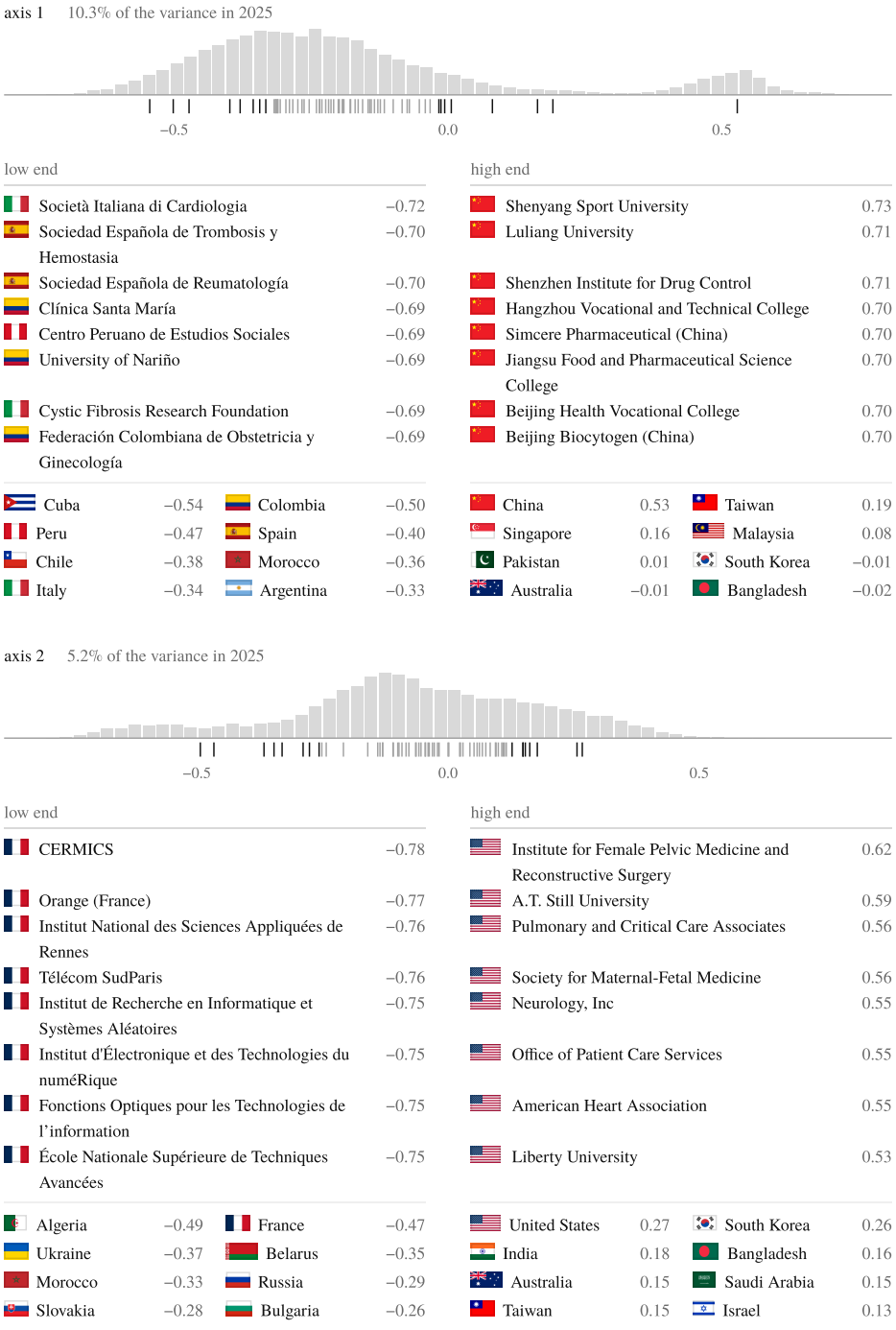


Figure S5. The eight institutions and the eight countries at each end of the leading axes in 2025, in the yearly analysis. Countries are ranked by their arrows among the 62 countries with a distance series, and the strip shows the scores of all 19,254 institutions, with one tick for each country, dark for the countries listed.

Table S20. Weighted R^2 of the institution scores on each description of the institutions, averaged over the years.

		Weighted R^2					
Analysis	Axis	Country	Partner countries	Fields	Type	Log weight	Place
Yearly	1	0.90	0.94	0.24	0.04	0.11	0.56
	2	0.64	0.70	0.28	0.10	0.14	0.28
	3	0.57	0.63	0.27	0.08	0.17	0.14
	4	0.66	0.69	0.12	0.07	0.01	0.13
	5	0.66	0.70	0.16	0.06	0.00	0.20
	6	0.61	0.64	0.16	0.06	0.01	0.13
	7	0.66	0.69	0.11	0.04	0.01	0.11
	8	0.71	0.75	0.08	0.03	0.00	0.08
Pooled	1	0.91	0.94	0.24	0.04	0.11	0.56
	2	0.62	0.68	0.28	0.08	0.17	0.30
	3	0.58	0.64	0.28	0.11	0.15	0.12
	4	0.66	0.69	0.12	0.07	0.01	0.14
	5	0.80	0.83	0.08	0.02	0.00	0.20
	6	0.38	0.40	0.33	0.15	0.00	0.16
	7	0.77	0.80	0.04	0.00	0.01	0.07
	8	0.77	0.80	0.06	0.02	0.00	0.10

S10.4 Closeness in shared author-years

For the 3,000 institutions with the largest weight, t_{ij} counts the author-years of one half that hold records at both i and j. With n_i the author-years of i and N those of the cell, independence expects $n_i n_j / N$. A Poisson regression fits the count on the cosine of i and j in the other half, in both directions and for every year,

$$\log \mathrm{E}[t_{ij}] = \log \frac{n_i n_j}{N} + \alpha + \beta \, (\cos_{ij} - \overline{\cos}),$$

where $\overline{\cos}$ is the mean cosine over the pairs. The measure is the factor $e^{0.1\beta}$, with β averaged over the two directions. Between two pairs of institutions whose cosines differ by 0.1, the closer pair has that many times the ratio of shared author-years to what independence expects. The factor stays between 2.12 and 2.22 in every year, and the cosine explains 62% to 76% of the deviance (Table S21).

Table S21. Shared author-years against cosine for the 3,000 institutions with the largest weight, by year. The factor $e^{0.1\beta}$ multiplies the shared author-years, relative to independence, for a cosine higher by 0.1.

Year	Factor for +0.1	Deviance explained (%)
2008	2.16	63.1
2009	2.15	62.9
2010	2.12	61.6
2011	2.16	64.2
2012	2.17	65.0
2013	2.15	64.5
2014	2.18	66.3
2015	2.19	67.6
2016	2.16	67.2
2017	2.17	68.3
2018	2.21	70.5
2019	2.21	71.9
2020	2.22	72.7
2021	2.21	71.5
2022	2.18	74.7
2023	2.14	71.1
2024	2.12	75.2
2025	2.14	76.4

S10.5 Distance to the rest on the axes

With $u = g_c - g_{\bar{c}}$ the gap between a country and the rest, projecting u on the orthonormal axes e_k of a map splits the square of distance to the rest exactly,

$$D_c^2 = \sum_k \frac{(u^\top e_k)^2}{S}.$$

For the eight countries of Figure 15, averaged over maps and years, the first eight axes hold 99% of D_c^2 for China and 95% for the United States (Table S22). For the other six they hold 27% to 73%, and the share stays below half for Türkiye, Saudi Arabia, Russia and Iran, whose distance lies mostly along the many smaller axes. Figure S6 places the institutions of China, the United States and the pair of the United States and India on the two axes that carry most of their arrows in 2025, and Section S12 splits both measures over institutions.

Table S22. Shares of the squared distance to the rest on the first eight axes and on the remaining axes, for the eight countries of Figure 15, averaged over maps and years.

	Share of D_c^2 (%)								
Country	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6	Axis 7	Axis 8	Rest
China	93.9	3.8	0.3	0.2	0.4	0.4	0.3	0.1	0.5
United States	0.5	64.3	5.5	10.5	5.0	5.4	2.6	1.1	5.1
Saudi Arabia	5.3	7.7	5.7	2.7	1.7	3.5	7.4	6.3	59.8
Russia	6.8	9.3	5.1	4.7	1.0	2.5	2.2	8.5	60.1
Türkiye	12.0	7.1	14.6	0.9	1.0	2.5	2.4	4.1	55.4
Germany	12.5	3.3	6.5	14.6	4.5	5.2	4.8	21.9	26.7
Iran	7.9	4.4	6.9	1.9	1.4	1.1	2.4	0.9	73.3
Spain	44.3	2.6	0.3	2.2	5.0	2.2	2.0	0.7	40.6

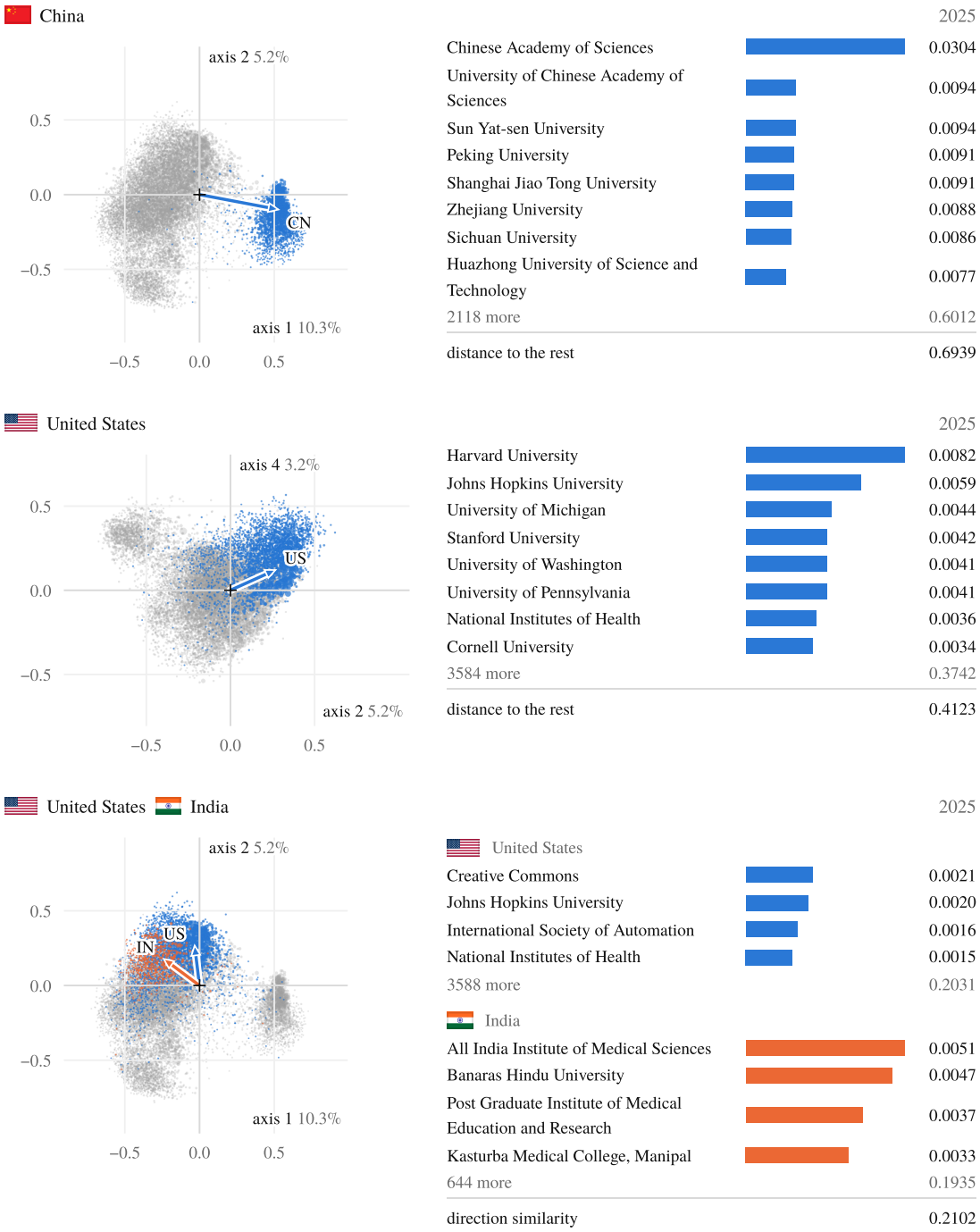


Figure S6. Institutions in 2025 on the two axes of the yearly map that carry most of a country's arrow, or of both arrows for a pair. The bars list the institutions that add most to the real distance to the rest, or to the real direction similarity for a pair.

S11 Clusters of countries

We group the 62 countries by how alike their movements are. In each half, a country's series is its centred net distance minus the mean of all countries in that half and year, which is own movement before the division by the noise sd. The similarity of two countries i and j is

$$r_{ij} = \frac{1}{2}\left[\operatorname{corr}_t\left(m^a_{i,t}, m^b_{j,t}\right) + \operatorname{corr}_t\left(m^b_{i,t}, m^a_{j,t}\right)\right],$$

where m^a and m^b are the series of the two halves and the correlations run over the 18 years from 2008 to 2025. Each correlation sets one half against the other, so its two sides carry independent noise, and the diagonal r_{ii}, a country's correlation with itself across halves, has a median of 0.58.

Average-linkage clustering on $1 - r$, with the leaves in optimal order, sorts the countries and cuts the tree into k clusters. To choose k, each half is clustered on the correlations of its own series, and the adjusted Rand index compares the two partitions for every k from two to twelve (Table S23). The halves agree best on three clusters, which hold 28, 31 and 3 countries, and on the site the reader can change the number of clusters in Figure 16 and Figure S7. The pairs go through the same procedure on their centred net direction similarity.

Table S23. Agreement of the two halves on the clusters of countries, by number of clusters k. Each half is clustered on its own series, and the adjusted Rand index compares the two partitions.

k	2	3	4	5	6	7	8	9	10	11	12
Adjusted Rand index	0.407	0.518	0.475	0.404	0.356	0.305	0.305	0.333	0.244	0.279	0.283

The mean path of a cluster averages the series of its members, and its noise band treats their noises as independent. Figure 16 shows these paths, Table S24 their change from 2008–2010 to 2023–2025 and Figure S7 the similarity r in cluster order.

Table S24. Clusters of countries. Mean r within is the mean similarity between distinct members. Change is the change of the mean path from 2008–2010 to 2023–2025, and change and noise sd are in units of the spread. The last column divides the change by its noise sd.

Cluster	Members	Mean r within	Change	Noise sd	In noise sd
Farther	28	0.253	+0.0194	0.0012	+15.9
Closer	31	0.198	−0.0183	0.0012	−15.6
Third	3	0.306	+0.0074	0.0038	+2.0

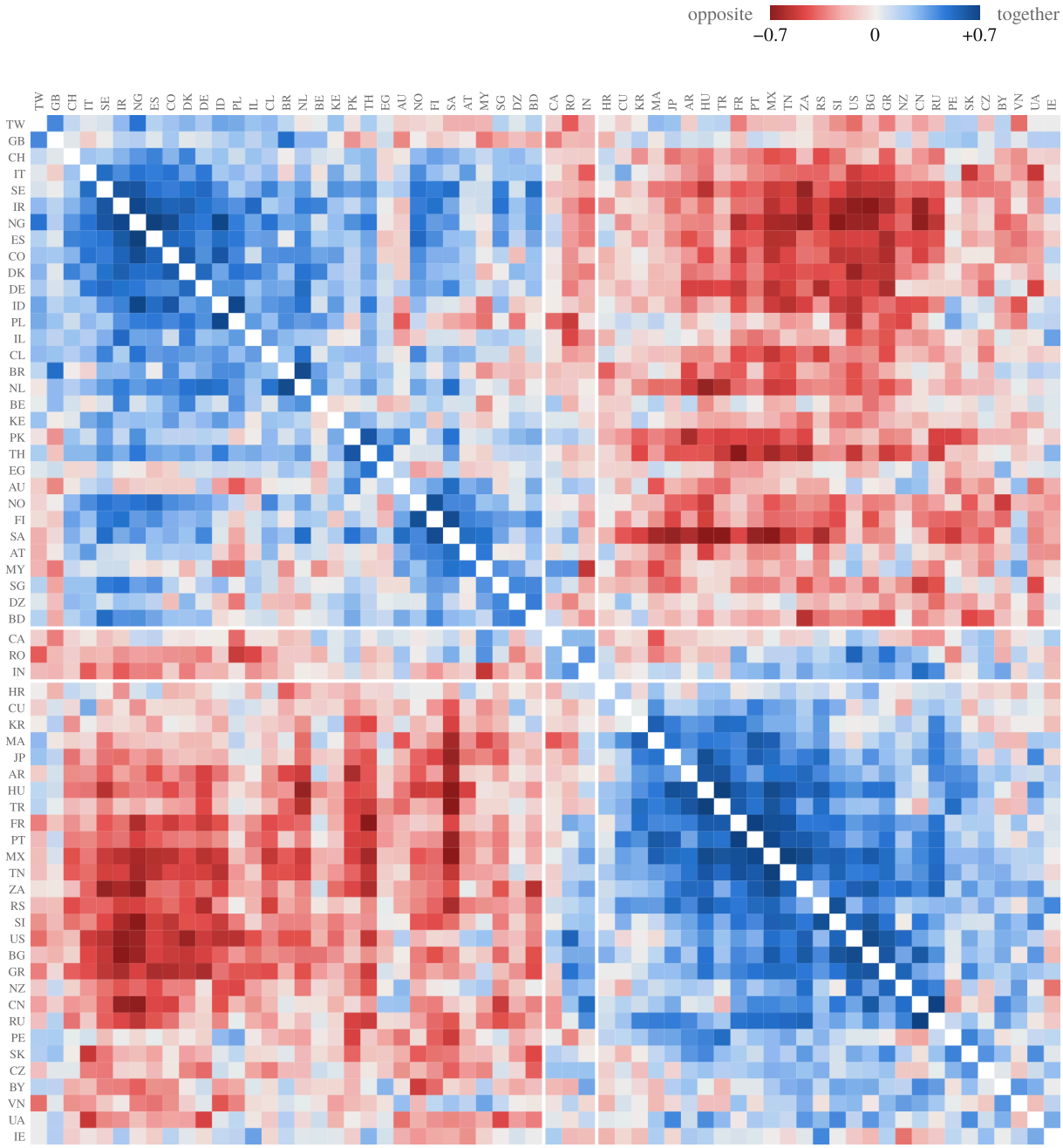


Figure S7. Similarity r of the 62 countries in cluster order. Blue cells mark countries that move together and red cells countries that move in opposite directions, with colours saturating at the ends of the key. White gaps separate the clusters.

S12 Institutions behind the movement

Both measures split exactly over institutions. With $\hat{u}$ the unit vector from $g_{\bar{c}}$ to g_c, a country's distance to the rest is a weighted sum over its institutions,

$$D_c = \sum_{i \in c} \frac{w_i}{W_c}\, p_i, \qquad p_i = \frac{(x_i - g_{\bar{c}}) \cdot \hat{u}}{\sqrt{S}},$$

so an institution adds much when it carries much of the country's weight and stands far from the rest along the line between the two centroids. With $\hat{b}$ the unit vector along the arrow of country b, the direction similarity of a and b splits over the institutions of a,

$$\cos(a, b) = \sum_{i \in a} \frac{w_i}{W_a}\, \frac{(x_i - m) \cdot \hat{b}}{\|g_a - m\|},$$

and in the same way over the institutions of b. Each contribution is averaged over seeds and halves like the series, and its net value subtracts the twin within each half. The change of a contribution is its mean net value over 2023–2025 minus its mean over 2008–2010, and the changes of a country's institutions sum to the change of its net distance to the rest. Half the difference between the changes of the two halves gives the noise, whose size follows the institution's own weight. Its root mean square within 16 bins of equal count by w_i gives $s(w_i)$, and the noise sd of a contribution's change is $(w_i/W_c)\, s(w_i)$.

Over all 62 countries, 4,939 of 18,581 institutions have a change beyond two noise sd. The United States holds 1,230 of these institutions. In the eight countries of Figure 15 these institutions carry from 43% of the change, for Iran, to 131%, for Germany, where the other institutions pull the other way. Table S25 gives, for each of these countries and each direction, up to five of the largest changes beyond two noise sd, and Figure S8 sets them against the rest of their country.

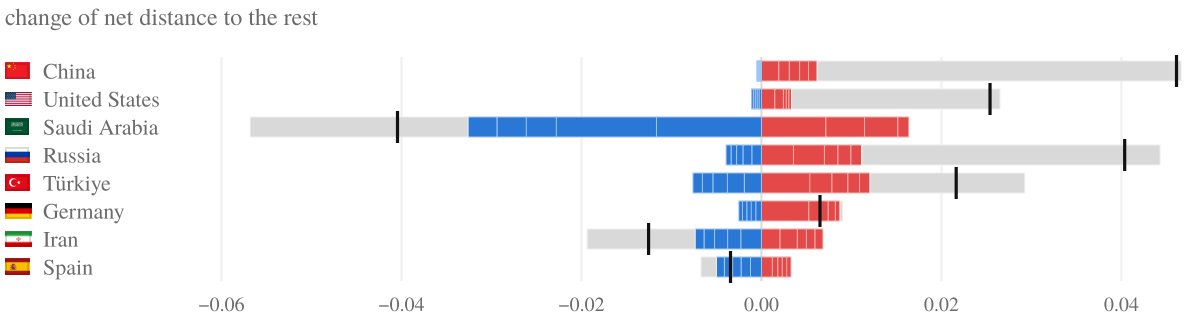


Figure S8. Change of each country's net distance to the rest from 2008–2010 to 2023–2025, split over its institutions, for the eight countries with the largest sums of squared own movement. Coloured segments are the largest changes beyond two noise sd, at most five in each direction, red where they move the country farther from the rest and blue where they move it closer. The grey segment sums all other institutions, and the black tick marks the country's change. Changes are in units of the spread.

Table S25. The largest changes beyond two noise sd in their country's net distance to the rest from 2008–2010 to 2023–2025, at most five in each direction, for the eight countries with the largest sums of squared own movement. The first row of a country gives the change of all its institutions together and the second that of all its institutions beyond two noise sd. Changes are in units of the spread. Institutions are as OpenAlex lists them, which includes some publishers and licence bodies.

Country	Institution	Change	In noise sd
China	All 2,126 institutions	+0.0461	
	764 beyond two noise sd	+0.0413	
	Chinese Academy of Sciences	+0.0019	+2.4
	Chinese Academy of Medical Sciences & Peking Union Medical College	+0.0012	+6.6
	Sichuan University	+0.0011	+4.9
	Sun Yat-sen University	+0.0010	+3.7
	Central South University	+0.0010	+4.7
	Zhejiang Ocean University	−0.0001	−7.0
	Ministry of Natural Resources	−0.0001	−4.0
	Southwest University	−0.0001	−2.1
	Nanjing Normal University	−0.0001	−3.4
	Harbin Engineering University	−0.0001	−2.3
United States	All 3,592 institutions	+0.0254	
	1,230 beyond two noise sd	+0.0188	
	Creative Commons	+0.0015	+28.9
	Boston Children's Hospital	+0.0010	+12.9
	Texas A&M University	+0.0003	+3.3
	AbbVie (United States)	+0.0003	+22.3
	Janssen (United States)	+0.0003	+17.1
	Washington University in St. Louis	−0.0003	−3.1
	International Society of Automation	−0.0002	−2.4
	Lawrence Berkeley National Laboratory	−0.0002	−3.6
	University of Arizona	−0.0002	−3.1
	Harvard University Press	−0.0002	−5.7
Saudi Arabia	All 42 institutions	−0.0404	
	24 beyond two noise sd	−0.0406	
	King Saud University	+0.0072	+2.6
	King Abdulaziz Medical City	+0.0043	+4.2
	King Faisal Specialist Hospital & Research Centre	+0.0037	+4.5
	Armed Forces Hospital	+0.0012	+2.9
	King Abdullah University of Science and Technology	−0.0117	−9.9
	King Abdulaziz University	−0.0111	−5.3
	Ministry of Education	−0.0033	−7.5
	King Khalid University	−0.0033	−3.5
	King Fahd Hospital of the University	−0.0032	−6.3
Russia	All 448 institutions	+0.0404	
	127 beyond two noise sd	+0.0381	
	Ministry of Health of the Russian Federation	+0.0036	+6.2
	Lomonosov Moscow State University	+0.0034	+4.4
	St Petersburg University	+0.0015	+4.5
	Don State Technical University	+0.0014	+10.2
	Institute for Theoretical and Experimental Physics	+0.0012	+8.3
	Institute of Catalysis and Petrochemistry	−0.0011	−8.9
	Siberian Branch of the Russian Academy of Sciences	−0.0010	−2.2
	Novosibirsk State University	−0.0007	−2.1
	Moscow Institute of Physics and Technology	−0.0006	−2.8
	Ural Branch of the Russian Academy of Sciences	−0.0005	−3.0
Türkiye	All 179 institutions	+0.0216	
	50 beyond two noise sd	+0.0158	
	Sağlık Bilimleri Üniversitesi	+0.0054	+3.7
	Ministry of Health	+0.0025	+4.8
	Anadolu University	+0.0018	+4.4
	Eskişehir City Hospital	+0.0013	+3.6
	Koç University	+0.0012	+4.4
	Cukurova University	−0.0019	−6.5
	Erciyes University	−0.0019	−5.4
	Süleyman Demirel Üniversitesi	−0.0016	−5.3
	Boğaziçi University	−0.0012	−4.1
	Middle East Technical University	−0.0010	−3.2

Table S25 (continued)

Country	Institution	Change	In noise sd
Germany	All 877 institutions	+0.0065	
	207 beyond two noise sd	+0.0085	
	Max Planck Society	+0.0053	+15.0
	German Cancer Research Center	+0.0013	+8.0
	German Center for Neurodegenerative Diseases	+0.0008	+14.0
	Charité - Universitätsmedizin Berlin	+0.0008	+2.2
	University of Lübeck	+0.0005	+2.9
	Friedrich-Alexander-Universität Erlangen-Nürnberg	−0.0006	−2.5
	University of Würzburg	−0.0005	−3.7
	Ruhr University Bochum	−0.0005	−2.9
	University of Bonn	−0.0005	−2.1
	Freie Universität Berlin	−0.0004	−2.1
Iran	All 174 institutions	−0.0125	
	55 beyond two noise sd	−0.0054	
	Shiraz University of Medical Sciences	+0.0021	+4.5
	Jahrom University of Medical Sciences	+0.0019	+7.1
	Kermanshah University of Medical Sciences	+0.0010	+3.6
	Mazandaran University of Medical Sciences	+0.0010	+4.0
	Isfahan University of Technology	+0.0009	+2.9
	University of Tehran	−0.0023	−2.6
	Tarbiat Modares University	−0.0015	−3.2
	Urmia University	−0.0014	−3.1
	Atomic Energy Organization of Iran	−0.0012	−5.5
	Soil Conservation and Watershed Management Research	−0.0010	−4.9
Spain	All 581 institutions	−0.0034	
	89 beyond two noise sd	−0.0029	
	Instituto de Salud Carlos III	+0.0012	+4.0
	Health Research Institute of the Balearic Islands	+0.0006	+10.1
	Instituto Murciano de Investigación Biosanitaria	+0.0005	+6.2
	Universidad de Valladolid	+0.0005	+3.2
	Universidad Internacional De La Rioja	+0.0005	+8.1
	Universitat de València	−0.0012	−3.0
	Universidad Autónoma de Madrid	−0.0010	−2.6
	Universitat Pompeu Fabra	−0.0009	−5.8
	Universidad de Granada	−0.0009	−3.0
	Instituto de Física Corpuscular	−0.0009	−9.4

For the twelve pairs of Figure 18, Figure S9 splits the net change of the direction similarity over the institutions of each side, and the methods site lists the institutions behind the level and the change of each pair.

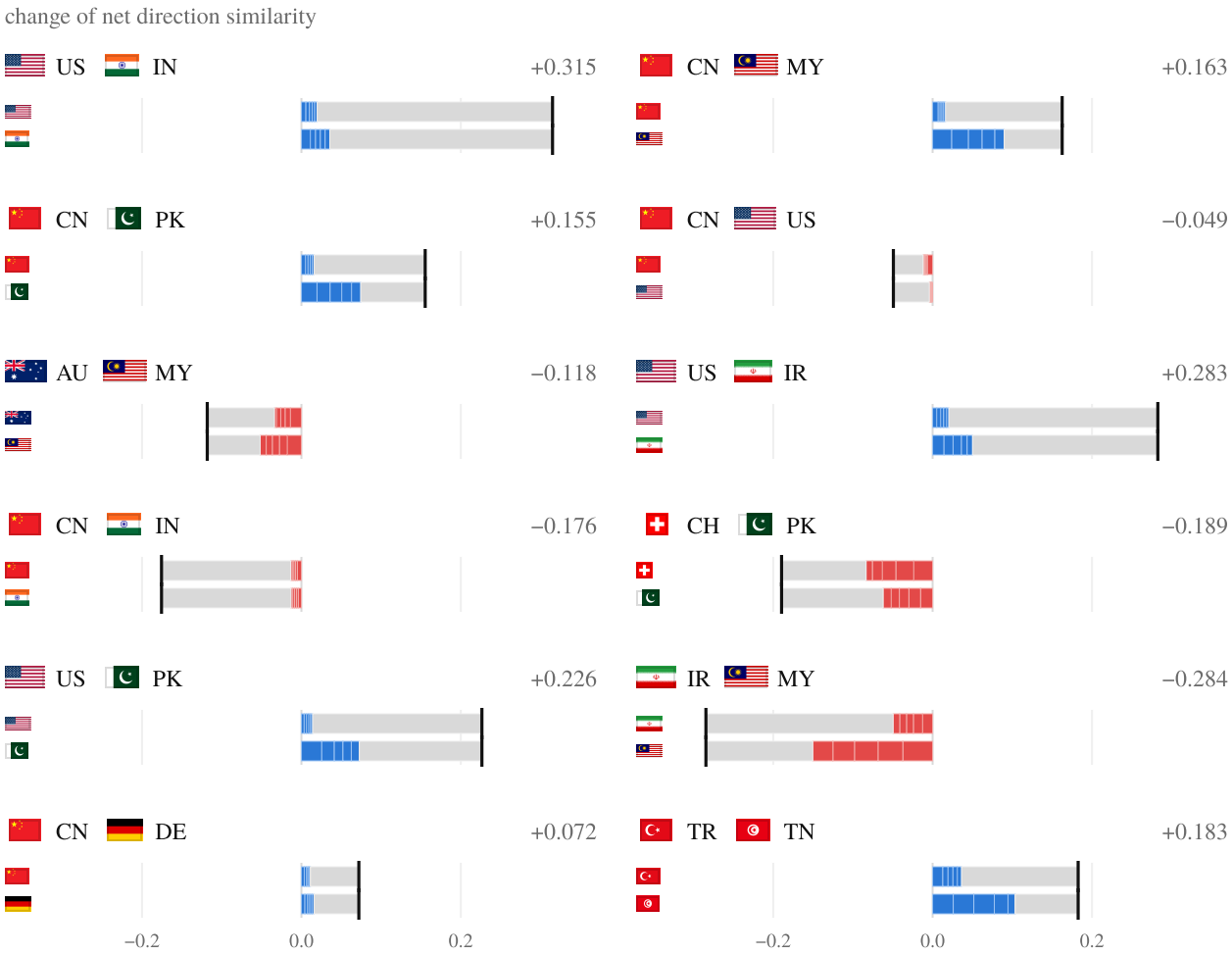


Figure S9. Net change of the direction similarity from 2008–2010 to 2023–2025, split over the institutions of each side, for the twelve pairs with the largest movement scores. Coloured segments are the five institutions of a side that add most in the direction of the pair's change, blue where they raise the similarity and red where they lower it. The grey segment sums all other institutions of that side. Both rows of a pair sum to the pair's change, marked by the black tick.

S13 Language regions and borders

Four countries hold parts that share a language with a neighbouring country, and the Franco-German border has two sides that face each other. Section 7 states the expectation and the result. Each institution takes its place from the city that OpenAlex gives it, joined to the GeoNames country files of 2026-11-08 (geonames.org, CC BY 4.0), and the parts follow the majority language of the canton, region or province.

- Switzerland by canton. Ticino forms the Italian part, Fribourg, Geneva, Jura, Neuchâtel, Vaud and Valais the French part, and the other cantons the German part.
- Belgium by region. Flanders forms the Dutch part and Wallonia the French part, and Brussels counts with the rest of Belgium.
- Canada by province. Quebec forms the French part and the other provinces and territories the rest.
- Italy by province. Bolzano forms South Tyrol against the rest of Italy.
- The Franco-German border. Moselle, Bas-Rhin and Haut-Rhin form the French side, and Saarland, Rhineland-Palatinate and the government regions of Karlsruhe and Freiburg the German side.

The rest of a part's country is every other institution of the cohort in that country, including the seven institutions whose city lies outside their country's file (Table S26).

Table S26. Institutions in each part and in the rest of its country, and the weight of each part as a share of the total weight of the cohort.

		Institutions		
Country	Part	In part	In rest of country	Weight (%)
Switzerland	German-speaking Switzerland	142	68	0.651
	French-speaking Switzerland	59	151	0.360
	Italian-speaking Switzerland	8	202	0.016
Belgium	Flanders	50	67	0.386
	Wallonia	16	101	0.111
Canada	Quebec	102	336	0.550
	Canada outside Quebec	334	104	1.780
Italy	South Tyrol	4	562	0.011
France	French side of the border	59	1,684	0.166
Germany	German side of the border	111	766	0.751

With g the weighted centroid of a group's institutions and m the centre of the world, the lean of a part p of country c towards a neighbour n is

$$L_{p,n} = \cos(g_p - m,\ g_n - m) - \cos(g_{c \setminus p} - m,\ g_n - m),$$

where $c \setminus p$ is the rest of c. A positive lean means the part points more towards the neighbour than the rest of its country does. Each part is set against the neighbours that share its language or lie across the border. Leans are averaged over seeds and halves like the series. Their noise sd follows the model of Section S6, with a scale from each lean's own half-differences and the yearly profile of the countries, and the average over years treats the yearly noise as independent.

Figure S10 shows each lean by year with its noise band, and Table S27 gives the averages. Every passing lean lies more than two noise sd above zero in at least 17 of the 18 years.

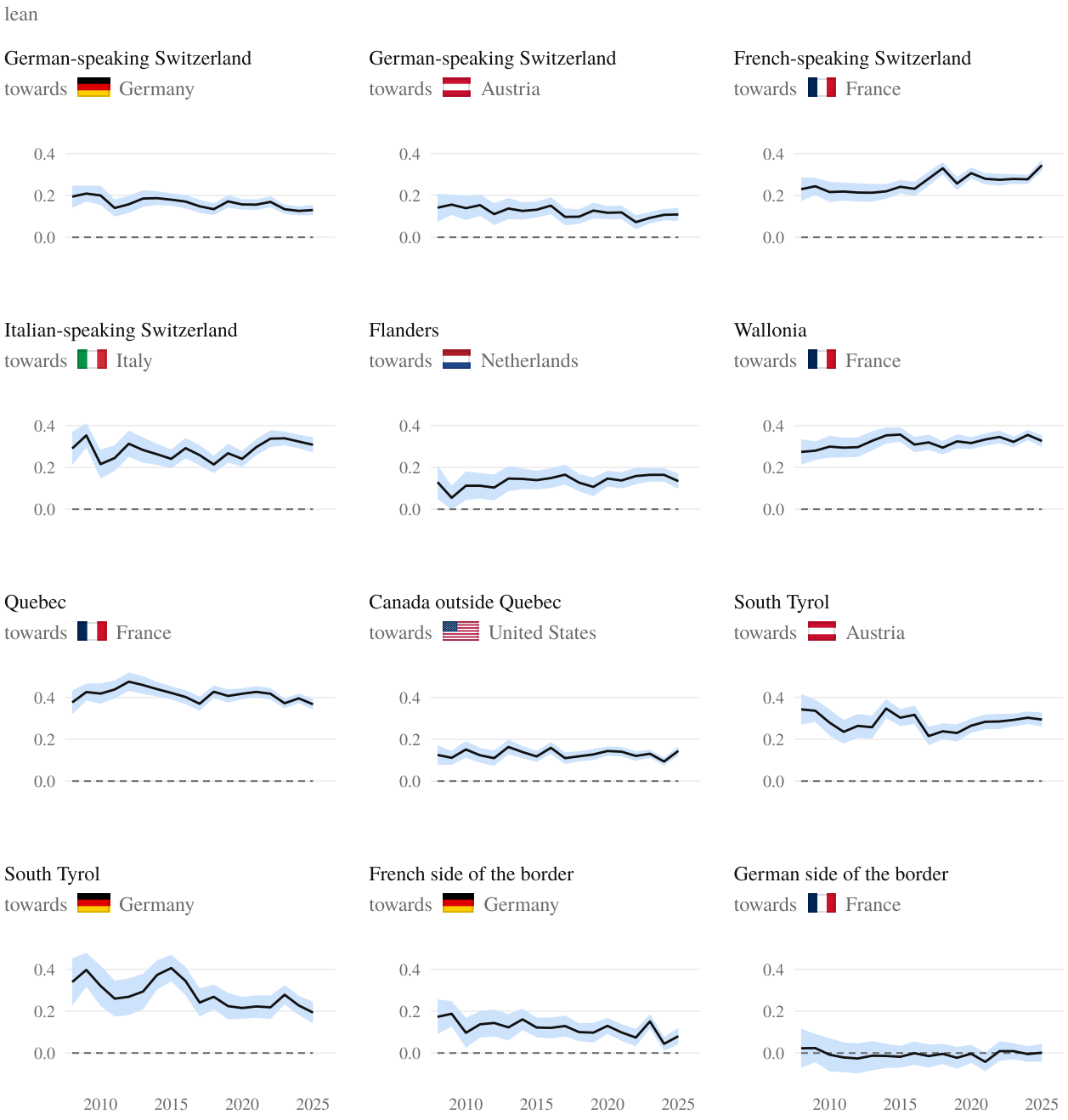


Figure S10. Lean of each part towards each neighbour by year, with its noise band. The dashed line marks zero, and all panels share one vertical axis.

Table S27. Lean of each part towards each neighbour, averaged over 2008–2025, with its noise sd. The last column counts the years whose lean lies more than two noise sd above zero.

Part	Neighbour	Mean lean	Noise sd	Years above two sd
German-speaking Switzerland	Germany	+0.164	0.004	18
	Austria	+0.121	0.005	18
French-speaking Switzerland	France	+0.259	0.004	18
Italian-speaking Switzerland	Italy	+0.282	0.006	18
Flanders	Netherlands	+0.133	0.006	17
Wallonia	France	+0.318	0.005	18
Quebec	France	+0.415	0.004	18
Canada outside Quebec	United States	+0.129	0.003	18
South Tyrol	Austria	+0.283	0.005	18
	Germany	+0.283	0.008	18
French side of the border	Germany	+0.120	0.006	18
German side of the border	France	−0.007	0.007	0

S14 Events

For Russia and Ukraine the series is the movement net of the common drift, in units of the spread, which is own movement before the division by the noise sd. For an event in year t_0, T_0 holds the three years before t_0 and T_1 the three years after, up to 2025. The change and its noise sd are

$$\Delta = \frac{1}{|T_1|}\sum_{t\in T_1} y_t - \frac{1}{|T_0|}\sum_{t\in T_0} y_t, \qquad \mathrm{sd}_\Delta^2 = \frac{1}{|T_1|^2}\sum_{t\in T_1} \mathrm{sd}_t^2 + \frac{1}{|T_0|^2}\sum_{t\in T_0} \mathrm{sd}_t^2,$$

where y_t is the series and sd_t its yearly noise sd. Table S28 gives the four changes, also in noise sd, and Figure S11 shows the series.

Table S28. Change around each event, for the common drift and, for Russia and Ukraine, the movement net of the drift. The change is the mean over the years after the onset minus the mean over the years before, in units of the spread, and its noise sd combines the yearly noise sd of both windows. Figure 20 multiplies each change by its expected sign, so values in the expected direction are positive there.

Event	Series	Onset	Before	After	Expected sign	Change	Noise sd	In noise sd
COVID-19	Common drift	2020	2017–2019	2021–2023	+	+0.0088	0.0006	+14.1
Reopening	Common drift	2023	2020–2022	2024–2025	−	−0.0010	0.0006	−1.6
Invasion of Ukraine	Russia	2022	2019–2021	2023–2025	+	+0.0171	0.0018	+9.4
Invasion of Ukraine	Ukraine	2022	2019–2021	2023–2025	−	−0.0088	0.0042	−2.1

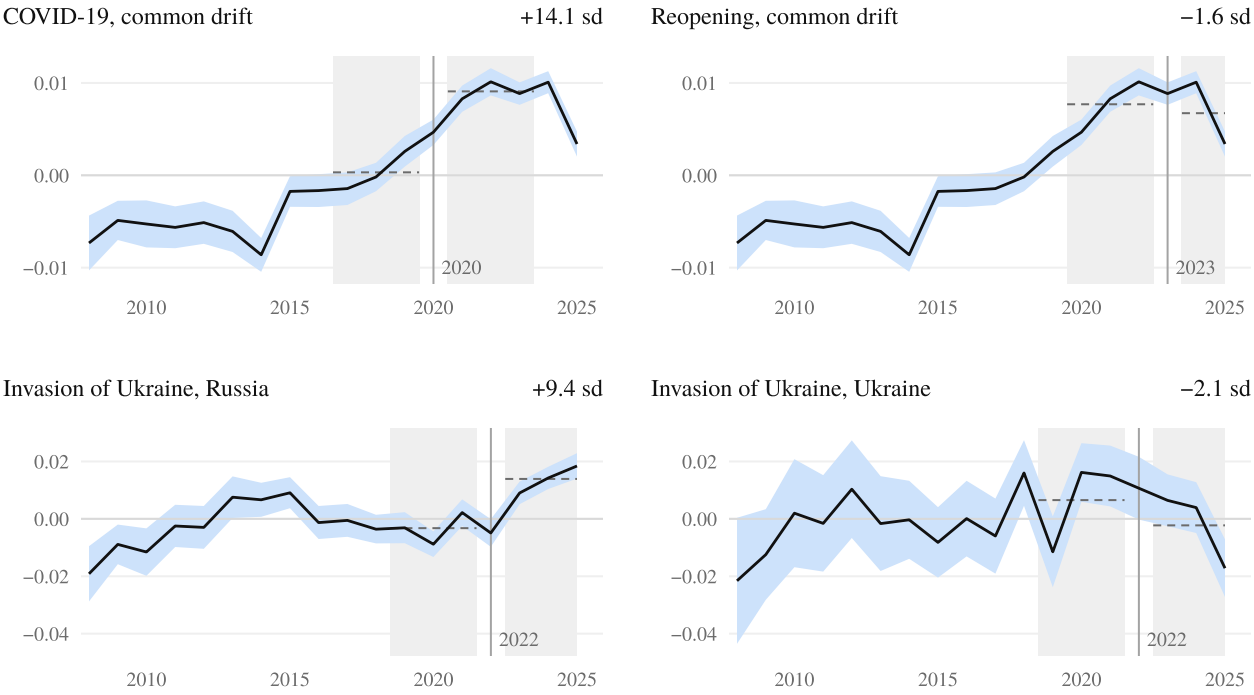


Figure S11. The series around each event, with their noise bands. The vertical line marks the onset, the shaded windows hold the years of the two means, and a horizontal line marks each mean. The two drift panels share one vertical axis, and so do the two country panels.

S15 Field views

A field view places each institution once in every field it works in. It reads a second world, trained like the main world, whose tokens join an institution to an OpenAlex subfield. Each record takes the subfield of one work its paper cites, drawn at random among the cited works that have a topic, so the subfield follows the literature the paper builds on. This world keeps the papers that cite a work with a topic, 84% of the records of the world built on primary topics. OpenAlex groups its subfields into fields, and twenty-six fields hold measured units.

Within a field, an institution's position is the average of its subfield vectors in that field, each scaled to unit length and weighted by the subfield's share of author-years over all cells. An institution enters a field when its subfields there hold at least three records in every cell of this world and of its twin, and its weight is its share of the field's author-years. Each field is then read like the main world, with its own twin, noise sd and common drift. Distance to the rest covers the countries with at least 20 institutions in the field, and direction similarity the pairs among the 40 countries with the largest weight there. With two countries the arrows point in opposite directions from the centre and their cosine is −1 in every year, so direction similarity needs three countries or more. The twenty-six fields hold 30,105 institution entries, 292 country series and 17,749 pair series (Table S29).

S15.1 The subfield of a record

A record can take its subfield from one work its paper cites or from the paper's primary topic, and we trained a world for each. The field views read the world whose centred net series agree better between the two halves, pooled over the fields and averaged over distance to the rest and direction similarity. Subfields drawn from the references reach 0.62 for distance and 0.26 for direction, and those of the primary topic reach 0.57 and 0.25, against 0.60 and 0.21 in the main world. On the 275 country series and 15,261 pair series that both worlds measure, the yearly real values correlate at 0.98 and 0.95.

S15.2 Results by field

Table S30 gives for each field the country farthest from the rest, the country with the most own movement and the pair with the highest movement score. Summed over the fields, the United States has the most own movement, 38% of it in

Engineering, and Brazil and Germany have the highest movement score, 6,500 over twenty-six fields, where pure noise gives about 442. In Nursing, Dentistry and Veterinary, the United States alone has a distance series, so its movement is the common drift of the field and its own movement is zero.

Table S29. Institutions, country series and pair series in each field, and the half agreement of the field views with subfields drawn from the references and with the primary topic. The counts refer to the world the field views read, and the last row pools the fields.

				References		Primary topic	
Field	Institutions	Countries	Pairs	Distance	Direction	Distance	Direction
Medicine	5,586	40	780	0.67	0.54	0.62	0.59
Biochemistry, Genetics and Molecular Biology	3,483	29	780	0.58	0.47	0.49	0.55
Engineering	2,772	21	780	0.70	0.40	0.56	0.49
Physics and Astronomy	1,704	18	780	0.77	0.55	0.76	0.53
Materials Science	1,649	16	780	0.78	0.67	0.69	0.48
Computer Science	1,598	16	780	0.70	0.52	0.62	0.38
Environmental Science	1,493	17	780	0.51	0.15	0.39	0.00
Agricultural and Biological Sciences	1,389	16	780	0.61	0.33	0.60	0.34
Neuroscience	1,251	13	780	0.67	0.34	0.69	0.27
Chemistry	1,228	13	780	0.83	0.49	0.74	0.54
Immunology and Microbiology	1,143	14	780	0.57	0.26	0.63	0.29
Social Sciences	960	11	780	0.22	0.08	0.49	0.11
Earth and Planetary Sciences	870	11	780	0.71	0.38	0.50	0.32
Mathematics	830	10	780	0.50	0.06	0.48	0.08
Psychology	793	10	780	0.10	0.11	0.45	0.08
Health Professions	689	7	780	0.25	0.08	0.16	0.09
Economics, Econometrics and Finance	503	9	741	0.05	0.09	0.40	0.11
Decision Sciences	398	6	630	0.16	0.08	0.40	0.00
Business, Management and Accounting	335	3	595	0.43	0.06	0.52	0.03
Pharmacology, Toxicology and Pharmaceutics	304	2	666	0.81	0.16	0.71	0.10
Nursing	242	1	666	0.74	0.12	0.50	0.14
Arts and Humanities	241	3	561	0.24	0.04	0.28	0.14
Energy	240	2	465	0.32	0.30	0.49	0.39
Dentistry	174	1	435	−0.08	0.09	0.14	−0.04
Chemical Engineering	161	2	300	0.55	0.13	0.01	0.12
Veterinary	69	1	210	0.40	0.03	0.44	0.02
All fields	30,105	292	17,749	0.62	0.26	0.57	0.25

S15.3 Subfields

The same rule applied to the subfields of OpenAlex measures units in 191 subfields, 122 of them with distance series (Table S31).

Table S30. The leading units of each field. The distance is the mean distance to the rest of the farthest country over 2008–2025. Own movement is summed as squares over the years, and the score is the movement score of the leading pair, where pure noise gives about 17. A field with one distance series has no own movement.

Field	Farthest from the rest	Distance	Most own movement	Leading pair	Score
Medicine	China	0.898	United States	Italy–Nigeria	792
Biochemistry	France	1.116	France	United Kingdom–India	479
Engineering	China	1.205	United States	Spain–Türkiye	1,308
Physics and Astronomy	China	1.003	Australia	Argentina–Singapore	461
Materials Science	China	1.262	Italy	Japan–Malaysia	1,163
Computer Science	China	1.193	United Kingdom	Taiwan–Iran	3,581
Environmental Science	China	1.292	China	United Kingdom–Germany	1,713
Agriculture and Biology	China	1.238	France	Serbia–Ireland	738
Neuroscience	France	1.191	United Kingdom	Italy–Spain	393
Chemistry	Russia	1.311	France	Iran–Austria	1,328
Immunology	France	1.150	China	Mexico–Portugal	242
Social Sciences	China	0.588	United States	Spain–Italy	92
Earth Sciences	China	1.354	Spain	Germany–Brazil	1,277
Mathematics	China	0.656	Germany	Japan–Greece	129
Psychology	France	0.458	Australia	Italy–Switzerland	93
Health Professions	United States	0.419	France	Switzerland–Singapore	102
Economics	China	0.910	France	Germany–Russia	96
Decision Sciences	China	0.905	China	Norway–Taiwan	128
Business	China	0.505	United States	Malaysia–Slovenia	81
Pharmacology	China	0.932	United States	Netherlands–India	180
Nursing	United States	0.427	–	Canada–Sweden	90
Arts and Humanities	France	0.372	United States	Finland–Mexico	81
Energy	China	1.186	United States	Australia–Canada	1,712
Dentistry	United States	0.342	–	Brazil–Taiwan	129
Chemical Engineering	China	1.011	United States	United States–Germany	476
Veterinary	United States	0.343	–	Belgium–Sweden	80

Table S31. Subfields with measured units, under the name of their field. Countries counts the distance series and pairs the direction series, the half agreement pools the units of each subfield, and the last column lists up to three countries with the most own movement, summed as squares over the years.

				Half agreement		
Subfield	Institutions	Countries	Pairs	Distance	Direction	Own movement
Medicine						
Surgery	1,607	16	780	0.56	0.37	ES, CN, GB
Epidemiology	1,334	14	780	0.59	0.44	CA, FR, CN
Pulmonary and Respiratory Medicine	1,299	14	780	0.75	0.28	US, CN, IT
Oncology	1,078	12	780	0.58	0.30	US, IT, JP
Cardiology and Cardiovascular Medicine	1,012	13	780	0.68	0.45	US, CN, TW
Physiology	991	13	780	0.51	0.18	FR, US, IT
Public Health, Environmental and Occupational Health	929	13	780	0.29	0.29	CA, ES, FR
Radiology, Nuclear Medicine and Imaging	735	9	780	0.60	0.29	IT, US, CN
Pathology and Forensic Medicine	706	11	780	0.62	0.31	US, IT, ES
Psychiatry and Mental health	630	11	630	0.35	0.23	AU, US, GB
Infectious Diseases	611	8	780	0.72	0.29	GB, IT, ES
Endocrinology, Diabetes and Metabolism	568	10	741	0.62	0.23	US, FR, CN
Neurology	567	8	630	0.53	0.17	US, FR, JP
Hematology	563	9	666	0.71	0.40	FR, US, CN
Rheumatology	500	8	595	0.66	0.27	US, IT, FR
Genetics	365	5	496	0.70	0.16	IT, FR, US
Pharmacology	361	3	666	0.61	0.15	US, DE, CN
Hepatology	330	6	378	0.61	0.15	FR, JP, IT
Nephrology	318	3	496	0.33	0.09	US, JP, CN
Pediatrics, Perinatology and Child Health	305	2	703	0.27	0.15	GB, US
Reproductive Medicine	231	2	496	0.50	0.17	US, CN
Ophthalmology	219	1	378	0.76	0.18	–
Obstetrics and Gynecology	210	1	351	0.50	0.01	–
Orthopedics and Sports Medicine	167	1	276	0.57	0.15	–

Table S31 (continued)

Subfield	Institutions	Countries	Pairs	Half agreement		Own movement
				Distance	Direction	
Emergency Medicine	166	1	253	0.71	0.08	–
Gastroenterology	123	1	300	0.39	0.07	–
Dermatology	113	1	231	−0.15	−0.04	–
Otorhinolaryngology	104	1	210	0.47	0.18	–
Immunology and Allergy	90	1	231	0.60	0.30	–
Biochemistry	84	1	210	0.09	0.10	–
Rehabilitation	74	1	136	−0.12	−0.01	–
Urology	74	1	153	0.58	0.16	–
Complementary and alternative medicine	70	1	153	0.36	0.05	–
Anesthesiology and Pain Medicine	69	1	105	0.34	0.17	–
Critical Care and Intensive Care Medicine	60	1	120	−0.13	0.05	–
Transplantation	42	0	55	–	0.26	–
Internal Medicine	40	1	55	0.78	0.20	–
Geriatrics and Gerontology	29	0	28	–	−0.03	–
Family Practice	12	0	3	–	−0.07	–
Biochemistry, Genetics and Molecular Biology						
Molecular Biology	2,483	20	780	0.59	0.47	FR, IN, RU
Genetics	1,129	12	780	0.47	0.24	IT, FR, US
Cancer Research	553	9	666	0.74	0.36	US, FR, CN
Cell Biology	419	5	595	0.58	0.37	FR, US, DE
Biophysics	96	1	210	0.36	0.23	–
Molecular Medicine	77	0	171	–	0.05	–
Clinical Biochemistry	61	0	120	–	0.08	–
Biotechnology	38	0	66	–	0.27	–
Biochemistry	25	0	28	–	0.27	–
Endocrinology	23	0	21	–	0.25	–
Structural Biology	13	0	21	–	0.13	–
Physiology	11	0	28	–	−0.03	–
Aging	10	0	10	–	−0.06	–
Engineering						
Electrical and Electronic Engineering	1,161	13	780	0.66	0.43	ES, IN, IT
Biomedical Engineering	947	13	780	0.61	0.56	US, ES, IT
Mechanical Engineering	426	5	780	0.80	0.35	JP, KR, FR
Computational Mechanics	304	3	528	0.88	0.54	FR, CN, US
Aerospace Engineering	300	4	561	0.68	0.24	FR, JP, US
Control and Systems Engineering	242	3	528	0.65	0.15	FR, CN, US
Mechanics of Materials	237	3	496	0.87	0.36	US, FR, CN
Civil and Structural Engineering	174	3	406	0.61	0.30	FR, CN, US
Building and Construction	52	0	171	–	−0.16	–
Ocean Engineering	48	1	171	0.75	0.55	–
Media Technology	22	0	15	–	0.32	–
Automotive Engineering	21	0	15	–	0.01	–
Industrial and Manufacturing Engineering	21	0	10	–	−0.12	–
Safety, Risk, Reliability and Quality	16	0	10	–	0.43	–
Physics and Astronomy						
Atomic and Molecular Physics, and Optics	823	9	780	0.63	0.42	ES, CN, IT
Nuclear and High Energy Physics	621	9	780	0.64	0.55	IT, DE, RU
Astronomy and Astrophysics	554	9	780	0.51	0.32	CN, DE, RU
Radiation	214	2	351	0.29	0.16	US, IT
Condensed Matter Physics	213	4	351	0.73	0.25	FR, CN, US
Statistical and Nonlinear Physics	110	2	171	0.26	0.18	US, CN
Instrumentation	37	0	55	–	0.35	–
Materials Science						
Materials Chemistry	1,159	11	780	0.76	0.69	IT, IN, US
Electronic, Optical and Magnetic Materials	424	5	780	0.82	0.72	FR, RU, US
Biomaterials	239	3	435	0.63	0.24	FR, CN, US
Polymers and Plastics	149	1	378	0.37	0.57	–
Surfaces, Coatings and Films	68	1	136	0.23	0.23	–
Ceramics and Composites	54	1	55	0.41	0.62	–
Metals and Alloys	8	0	6	–	−0.04	–
General Materials Science	4	0	3	–	0.41	–

Table S31 (continued)

Subfield	Institutions	Countries	Pairs	Half agreement		Own movement
				Distance	Direction	
Computer Science						
Artificial Intelligence	677	10	780	0.61	0.48	CN, DE, IT
Computer Vision and Pattern Recognition	406	4	666	0.51	0.35	FR, DE, US
Computer Networks and Communications	297	3	630	0.75	0.57	FR, CN, US
Computational Theory and Mathematics	217	3	666	0.61	0.17	US, FR, CN
Information Systems	207	2	528	0.81	0.18	US, CN
Hardware and Architecture	73	1	78	0.13	0.20	–
Signal Processing	54	1	91	0.27	0.27	–
Human-Computer Interaction	12	0	15	–	−0.09	–
Computer Graphics and Computer-Aided Design	11	0	10	–	0.33	–
Environmental Science						
Ecology	415	5	741	0.67	0.15	GB, FR, CA
Global and Planetary Change	387	4	703	0.37	0.21	GB, FR, CN
Health, Toxicology and Mutagenesis	234	2	465	0.44	0.20	US, CN
Water Science and Technology	152	1	378	0.00	−0.17	–
Nature and Landscape Conservation	136	1	300	0.36	0.15	–
Pollution	95	1	325	0.60	0.04	–
Environmental Engineering	90	2	136	0.14	0.24	US, CN
Environmental Chemistry	76	1	190	0.30	0.51	–
Management, Monitoring, Policy and Law	19	0	45	–	0.10	–
Waste Management and Disposal	16	0	15	–	0.21	–
Ecological Modeling	3	0	3	–	0.63	–
Agricultural and Biological Sciences						
Plant Science	696	8	780	0.43	0.20	JP, FR, DE
Ecology, Evolution, Behavior and Systematics	183	2	435	0.22	0.10	US, FR
Food Science	167	3	496	0.55	0.29	FR, US, CN
Soil Science	104	2	190	0.11	0.37	CN, US
Agronomy and Crop Science	60	0	91	–	0.06	–
Insect Science	57	0	120	–	0.35	–
Animal Science and Zoology	49	0	66	–	0.18	–
Aquatic Science	27	0	66	–	0.34	–
General Agricultural and Biological Sciences	7	0	6	–	0.07	–
Forestry	5	0	6	–	−0.10	–
Neuroscience						
Cognitive Neuroscience	546	9	595	0.56	0.12	FR, GB, CN
Cellular and Molecular Neuroscience	458	7	703	0.70	0.37	DE, IT, GB
Neurology	160	1	231	0.47	0.11	–
Endocrine and Autonomic Systems	80	1	120	0.37	0.21	–
Sensory Systems	63	1	153	0.53	0.26	–
Developmental Neuroscience	59	1	105	0.21	0.26	–
Behavioral Neuroscience	18	0	28	–	0.03	–
Chemistry						
Organic Chemistry	631	10	780	0.83	0.62	RU, IT, GB
Spectroscopy	292	4	406	0.78	0.31	FR, CN, US
Inorganic Chemistry	219	3	351	0.82	0.47	FR, US, CN
Physical and Theoretical Chemistry	87	1	190	0.43	0.51	–
Analytical Chemistry	56	1	91	0.77	0.35	–
Electrochemistry	3	0	3	–	0.41	–
Immunology and Microbiology						
Immunology	825	12	780	0.38	0.19	ES, IT, CN
Virology	120	1	153	0.51	0.43	–
Parasitology	73	1	171	0.23	0.17	–
Microbiology	61	1	120	0.36	0.01	–

Table S31 (continued)

Subfield	Institutions	Countries	Pairs	Half agreement		Own movement
				Distance	Direction	
Social Sciences						
Sociology and Political Science	379	5	630	0.23	0.05	US, CA, GB
Education	107	1	171	0.36	0.13	–
Political Science and International Relations	87	1	190	0.06	0.03	–
Health	76	1	45	0.64	0.08	–
Demography	22	0	36	–	0.06	–
Anthropology	21	0	28	–	−0.02	–
Safety Research	21	0	28	–	0.09	–
Gender Studies	19	0	15	–	0.40	–
Transportation	6	0	6	–	−0.18	–
Law	3	0	3	–	0.08	–
Earth and Planetary Sciences						
Atmospheric Science	322	4	595	0.54	0.22	GB, FR, US
Geophysics	215	4	325	0.50	0.30	CN, FR, US
Oceanography	192	2	406	0.49	0.18	US, FR
Paleontology	86	0	190	–	0.33	–
Earth-Surface Processes	39	0	105	–	0.15	–
Geochemistry and Petrology	27	0	55	–	0.13	–
Geology	25	0	45	–	0.23	–
Mathematics						
Statistics and Probability	397	5	595	0.26	0.05	US, FR, GB
Applied Mathematics	79	1	153	0.64	0.15	–
Mathematical Physics	75	1	171	0.73	0.20	–
Geometry and Topology	47	0	105	–	0.06	–
Numerical Analysis	8	0	10	–	0.18	–
Modeling and Simulation	5	0	6	–	0.16	–
Psychology						
Clinical Psychology	330	4	378	−0.11	0.07	DE, GB, CA
Social Psychology	208	2	351	0.19	0.06	NL, US
Experimental and Cognitive Psychology	196	1	325	0.02	0.03	–
Developmental and Educational Psychology	62	1	78	0.18	0.04	–
Applied Psychology	21	0	10	–	0.20	–
Health Professions						
General Health Professions	301	4	276	0.23	0.17	AU, US, GB
Health Information Management	94	1	45	0.29	0.07	–
Emergency Medical Services	76	1	105	0.22	0.12	–
Physical Therapy, Sports Therapy and Rehabilitation	31	0	36	–	−0.07	–
Occupational Therapy	7	0	6	–	0.18	–
Economics, Econometrics and Finance						
Economics and Econometrics	330	4	528	0.24	0.06	FR, NL, US
Finance	86	1	120	0.22	0.05	–
General Economics, Econometrics and Finance	16	0	10	–	0.06	–
Decision Sciences						
Management Science and Operations Research	152	2	325	0.32	−0.09	US, CN
Statistics, Probability and Uncertainty	117	1	210	0.00	0.03	–
General Decision Sciences	8	0	3	–	−0.12	–
Information Systems and Management	8	0	10	–	0.12	–
Business, Management and Accounting						
Strategy and Management	30	0	91	–	−0.10	–
Accounting	27	0	45	–	0.03	–
Organizational Behavior and Human Resource Management	12	0	21	–	0.09	–
Management Information Systems	7	0	15	–	0.25	–
Management of Technology and Innovation	3	0	3	–	0.08	–
Pharmacology, Toxicology and Pharmaceutics						
Pharmacology	119	1	325	0.61	0.24	–
Pharmaceutical Science	44	0	120	–	0.42	–
Toxicology	17	0	21	–	0.31	–

Table S31 (continued)

Subfield	Institutions	Countries	Pairs	Half agreement		Own movement
				Distance	Direction	
Nursing						
Nutrition and Dietetics	217	1	630	0.76	0.12	–
Arts and Humanities						
Archeology	15	0	28	–	0.23	–
Philosophy	12	0	15	–	0.18	–
History and Philosophy of Science	7	0	21	–	−0.02	–
Language and Linguistics	5	0	3	–	−0.29	–
History	4	0	3	–	0.09	–
Energy						
Renewable Energy, Sustainability and the Environment	223	2	465	0.31	0.32	US, CN
Dentistry						
Periodontics	49	0	91	–	−0.09	–
Oral Surgery	40	0	105	–	0.18	–
Orthodontics	34	0	91	–	0.18	–
Chemical Engineering						
Catalysis	62	1	91	0.31	0.40	–
Fluid Flow and Transfer Processes	22	0	21	–	0.66	–
Bioengineering	3	0	3	–	−0.35	–
Filtration and Separation	3	0	3	–	0.01	–
Veterinary						
Small Animals	53	1	120	0.23	0.08	–